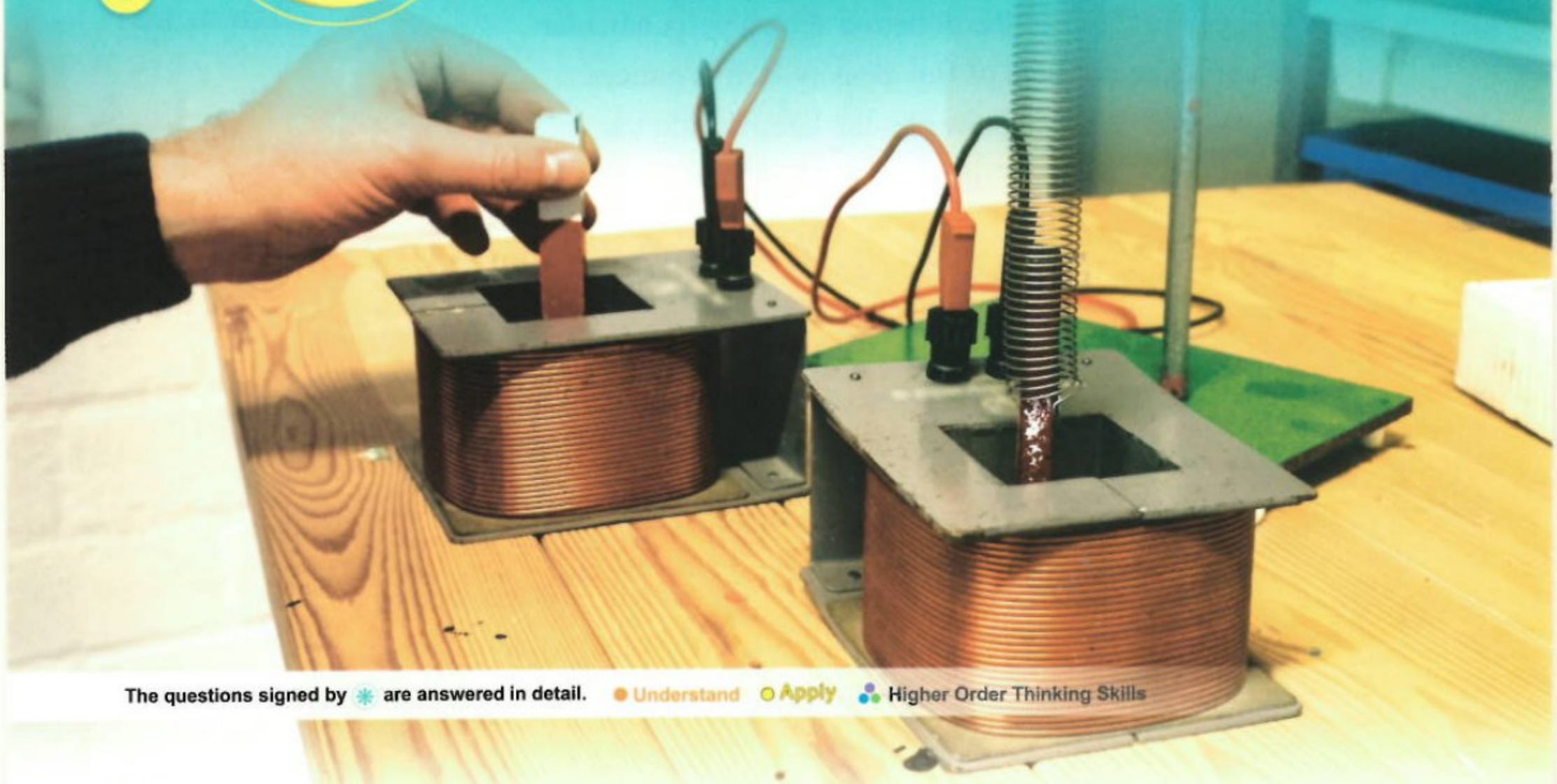
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Chapter 3

Questions on Lesson Two

- Mutual Induction between Two Coils.
- Self Induction of the Coil.

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

## Important guidelines

**⇒ The mutual induction between two coils:**

- To determine the induced emf generated in the secondary coil by mutual induction $(emf)_2$:

$$(emf)_2 = - M \frac{\Delta I_1}{\Delta t} = - N_2 \frac{\Delta(\phi_m)_2}{\Delta t}$$

(Where: $\Delta I_1$ is the change in the electric current intensity passing in the primary coil, $\Delta t$ is the time of that change)

- To determine the coefficient of mutual induction between two coils (M):

$$M = \frac{\mu N_1 N_2 A}{\ell} \qquad M = \frac{(emf)_2}{\Delta I_1 / \Delta t}$$

- If the time ($\Delta t$) is not given: $M \Delta I_1 = N_2 \Delta (\phi_m)_2$

**⇒ The self induction of a coil:**

- To determine the emf generated in the solenoid by self induction:

$$emf = - L \frac{\Delta I}{\Delta t} = - N \frac{\Delta(\phi_m)}{\Delta t}$$

(Where: $\frac{\Delta I}{\Delta t}$ is the rate of change in the electric current intensity passing in the coil)



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- To determine the coefficient of self induction for a solenoid (L):

$$L = \frac{\mu A N^2}{\ell} \qquad L = \frac{emf}{\Delta I/\Delta t}$$

- If the time ($\Delta t$) is not given:

$$L\Delta I = N\Delta\phi_m$$

- To compare between the coefficients of self induction for two solenoid coils in the same medium:

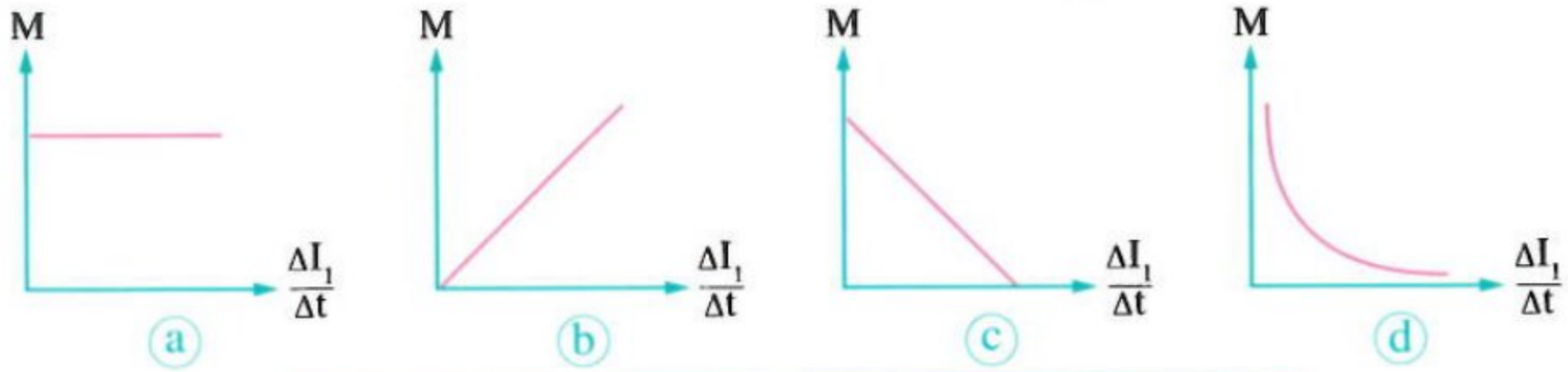
$$\frac{L_1}{L_2} = \frac{A_1 N_1^2 \ell_2}{A_2 N_2^2 \ell_1} = \frac{r_1^2 N_1^2 \ell_2}{r_2^2 N_2^2 \ell_1}$$

## First Multiple choice questions

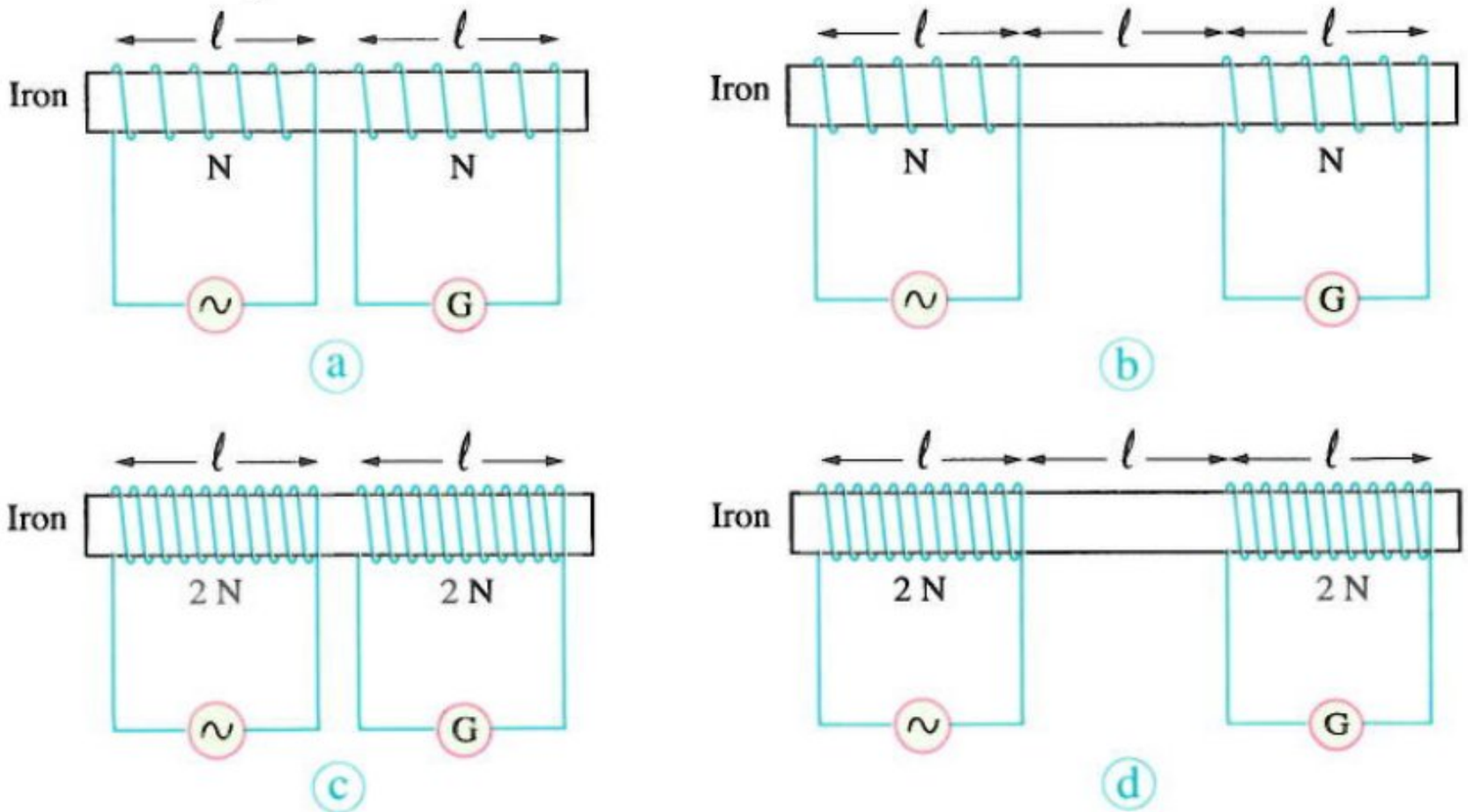
Using the following constant if it is needed: $\mu_{air} = 4\pi \times 10^{-7}$ Wb/A.m

### The mutual induction between two coils

1 Which of the following graphs represents the relation between the mutual induction coefficient (M) between two coils and the time rate of change in the intensity of the electric current which is passing in the primary coil $\left(\frac{\Delta I_1}{\Delta t}\right)$?




2 Which of the following pairs of coils has the greatest mutual induction coefficient, given that all coils have equal areas?






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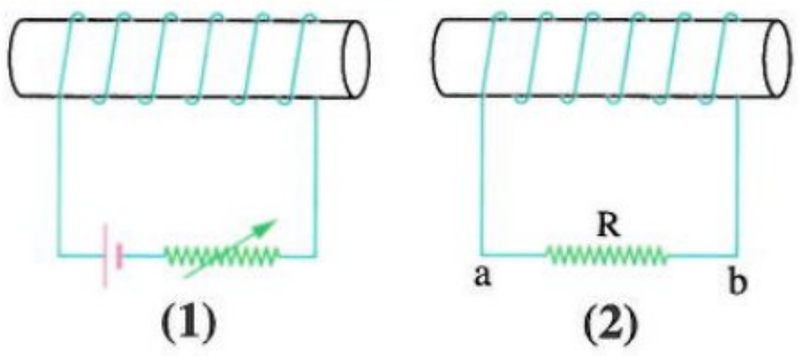
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3 A primary coil was connected to a DC source while being placed inside a secondary coil, so when opening the circuit of the primary coil, the generated current in the circuit of the secondary coil will be ..... .

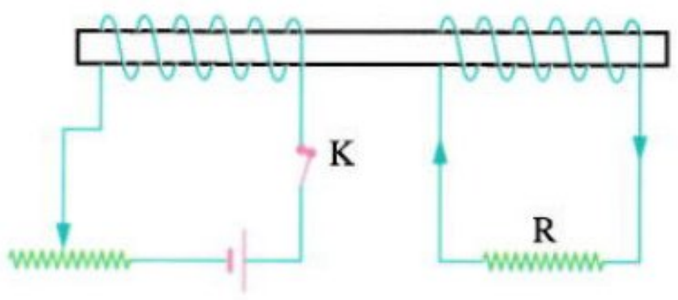
(a) forward induced instantaneous current
(b) back induced instantaneous current
(c) alternating current
(d) direct current

4 In the opposite figure, an induced electric current will be generated in circuit (2) and passes from point a to point b through resistor R when ..... .



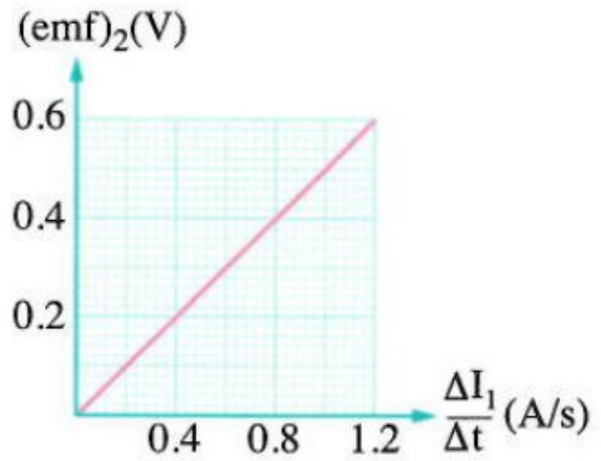

(a) moving the two circuits together at the same speed to the right side
(b) moving one of the circuits towards the other
(c) increasing the variable resistance in circuit (1)
(d) removing the iron core from one of the two circuits

5 An induced electric current will be generated in the secondary coil in a direction as shown in the opposite circuit, at ..... .



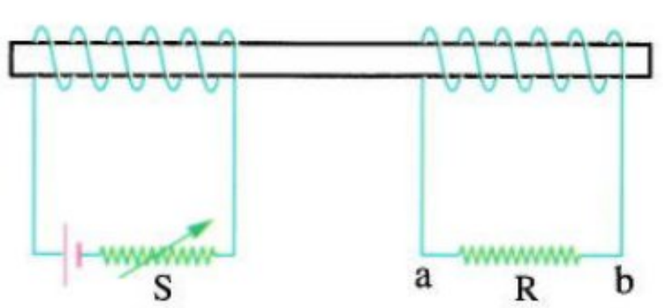

(a) opening switch K
(b) decreasing the variable resistance
(c) increasing the variable resistance
(d) moving the two coils away from each other

6 In an experiment for studying the mutual induction between two coils (1, 2), the relation between the magnitude of the induced emf $(emf)_2$ in coil (2) and the rate of change of the current $\left(\frac{\Delta I_1}{\Delta t}\right)$ in coil (1) was as shown in the opposite graph, so the mutual inductance between the two coils is ..... .




(a) 0.1 H (b) 0.2 H (c) 0.4 H (d) 0.5 H

7 In the opposite figure, during increasing the variable resistor (S), the potential of point (a) becomes ..... that of point (b).

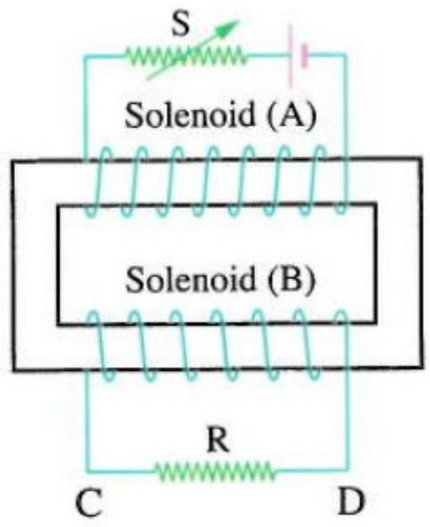



(a) greater than (b) less than
(c) equal to
(d) we can't determine the answer unless by knowing the value of resistance R



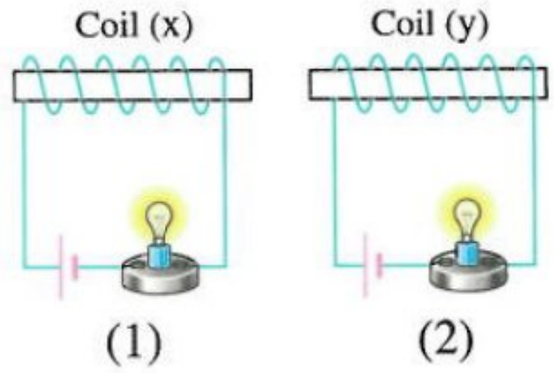
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**8** Two adjacent solenoids (A), (B) where solenoid (A) is connected to an electric circuit and solenoid (B) is connected to a resistor (R) as in the opposite figure. When value of the variable resistor (S) is decreased in the circuit of solenoid (A), the electric potential at point (C) becomes ..... .



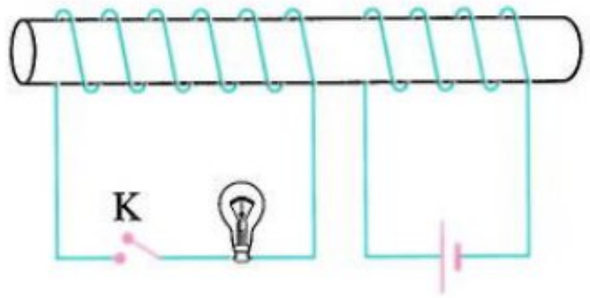

(a) greater than the electric potential at point (D)
(b) less than the electric potential at point (D)
(c) equals the electric potential at point (D)
(d) indeterminable.

**9** Two coaxial solenoids each carries electric current I and connected to two bulbs (1), (2) as in the opposite figure, when moving the solenoid (x) with a uniform velocity away from solenoid (y), the luminosity of ..... .



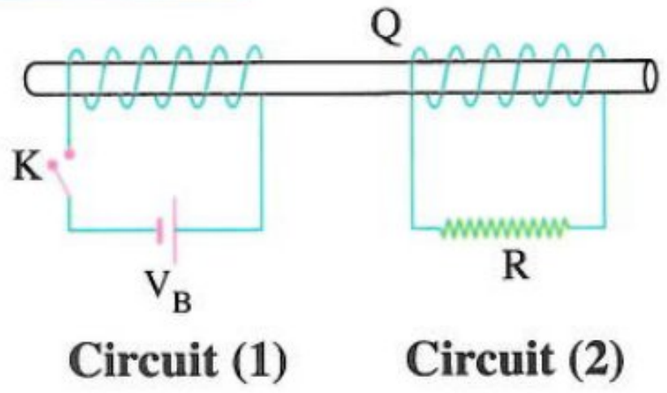

| | Bulb (1) | Bulb (2) |
|---|---|---|
| (a) | increases | increases |
| (b) | decreases | increases |
| (c) | increases | decreases |
| (d) | decreases | decreases |

**10** An isolated primary and secondary solenoids are coiled around an iron cylinder; the primary solenoid is connected to a battery while the secondary is connected to an electric bulb and a switch (K) as in the opposite figure. When switch (K) is closed the bulb ..... .




(a) illuminates momentarily then gets off (b) illuminates and doesn't get off
(c) illuminates and its luminosity decreases gradually till it gets off
(d) does not illuminate at all

**11** In the opposite figure, at the moment of closing circuit (1), then in circuit (2) ..... .



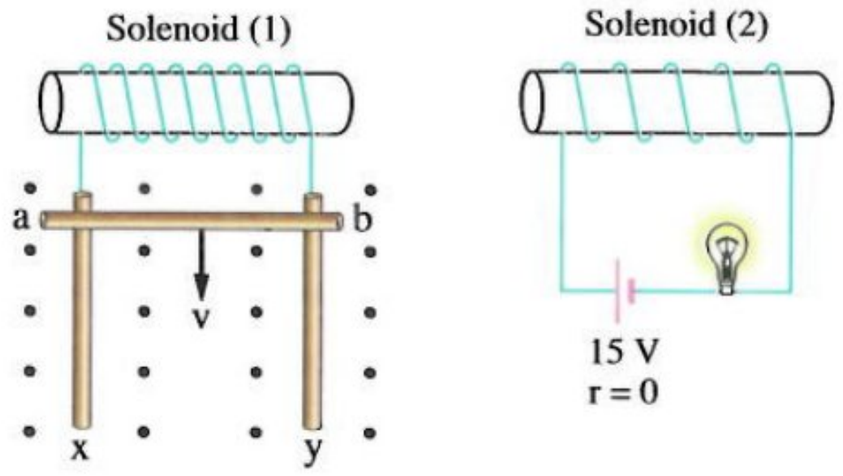

| | The terminal Q | The direction of the current |
|---|---|---|
| (a) | south | in the same current direction as circuit (1) |
| (b) | north | in the same current direction as circuit (1) |
| (c) | south | in opposite direction to the current of circuit (1) |
| (d) | north | in opposite direction to the current of circuit (1) |



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12 In the opposite figure, two adjacent solenoids (1), (2), where solenoid (1) is connected to two metal rails (x), (y) having the same resistance R. A metal rod moves while touching them with a uniform velocity (v) perpendicular to a uniform magnetic field, so if the magnitude of the induced emf in the rod (ab) equals 10 V, the luminosity of the bulb which is connected to solenoid (2) during the movement of the rod ..... .




(a) increases (b) decreases
(c) doesn't change (d) the answer is indeterminable

13 ✳ The mutual inductance between two coils is 0.1 H. If the intensity of current has dropped in the primary from 4 A to zero within 0.01 s, the induced emf across the secondary coil is equal to ..... .

(a) 20 V (b) 40 V (c) 60 V (d) 80 V

14 ✳ A primary solenoid is 10 cm long, consists of 200 turns and carries an electric current of 4 A, it has a soft iron core of permeability 0.002 Wb/A.m, on which a secondary coil of diameter 3.5 cm is wound with $10^5$ turns if the current is cut off from the primary coil within 0.01 s, then:

(i) The generated emf in the secondary coil equals ..... .

(a) $1.54 \times 10^5$ V (b) $4.32 \times 10^5$ V (c) $6.21 \times 10^6$ V (d) $8.8 \times 10^6$ V

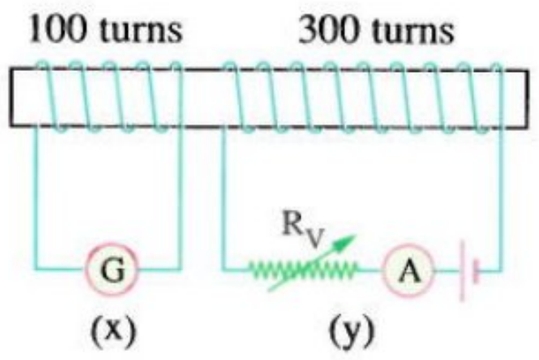
(ii) The mutual induction coefficient between the two coils is ..... .

(a) 130 H (b) 193 H (c) 385 H (d) 522 H

15 ✳ Two coils X and Y are adjacent. The number of turns of Y is 2000 turns, if a current of intensity 7 A flows in coil X and produced a magnetic flux of $2.5 \times 10^{-4}$ Wb through coil Y, then the mutual induction coefficient between the two coils equals ..... .

(a) 0.01 H (b) 0.03 H (c) 0.05 H (d) 0.07 H

16 ✳ The opposite figure represents two adjacent solenoids, the mutual inductance between them is 0.01 H. If the current intensity in coil y has changed by a value of $\Delta I$, the magnetic flux that cuts coil x changes by a value of $2 \times 10^{-3}$ Wb through the same time interval, then the value of the change in the current intensity ($\Delta I$) in the coil y is ..... .

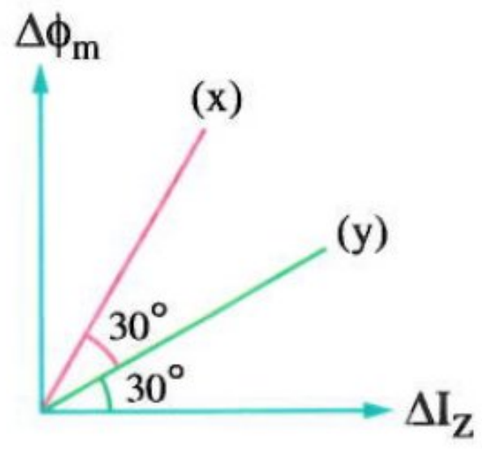



(a) 2 A (b) 5 A (c) 10 A (d) 20 A



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17 Two secondary solenoids (x), (y) of number of turns N, 2 N respectively, each is placed individually besides a primary solenoid (z) such that they have common axis every time. The opposite graph represents the relation between the change in the magnetic flux ($\Delta\phi_m$) and the change in the current ($\Delta I_z$) passing in the primary solenoid, so the ratio between the mutual induction coefficient in each case $\left(\frac{M_{xz}}{M_{yz}}\right)$ equals ..... .

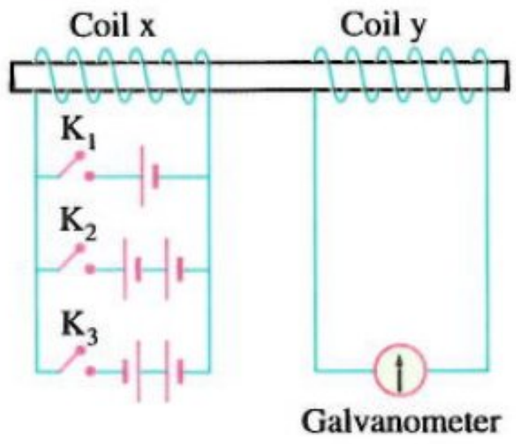



(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) 3

18 An electric current of intensity 10 A passes through one of two adjacent coils between which the mutual inductance is 0.3 H, when this current vanishes to zero, an induced emf of 60 V is generated in the other coil, then the decay time of the current in the first coil is equal to ..... .

(a) 0.005 s (b) 0.05 s (c) 0.04 s (d) 0.4 s

19 The opposite figure shows two similar coils x, y, each of resistance R. Coil x is connected through switches $K_1$, $K_2$ and $K_3$ to similar electric cells of negligible internal resistances. At the moment of closing switch $K_1$, the pointer of the galvanometer which is connected to coil y gets deflected by an angle θ, then the deflection angle of the galvanometer pointer at the moment of ..... .

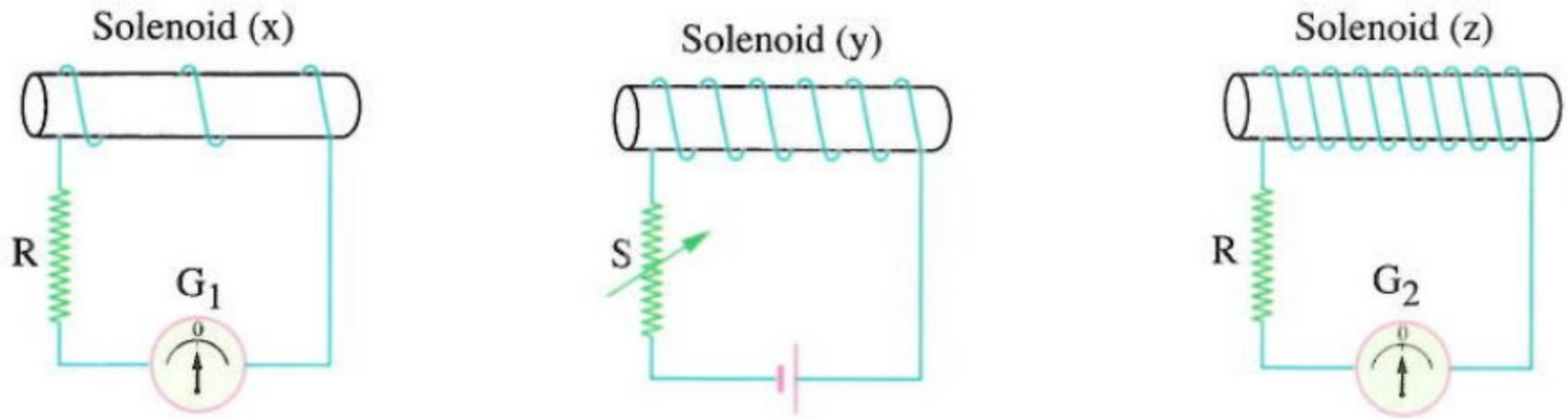



| | Closing switch $K_2$ only | Closing switch $K_3$ only |
|---|---|---|
| (a) | greater than (θ) | zero |
| (b) | greater than (θ) | greater than (θ) |
| (c) | equals (θ) | zero |
| (d) | less than (θ) | less than (θ) |

20 Three adjacent solenoids (x), (y), (z) of negligible resistances having the same common axis, same length and their numbers of turns are N, 2 N, 3 N respectively. Each of the solenoids (x), (z) is connected to a resistor (R) and a galvanometer which its zero is in the middle of its scale and a resistor (R). The solenoid (y) is connected to a variable resistor (S) and a battery as shown in the following figure;



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When the resistance of the variable resistor (S) is reduced, the pointer of the galvanometer ($G_1$) is deflected by an angle ($\theta$) to the right side of the zero scale, so the galvanometer's pointer ($G_2$) deflects by an angle ..... .

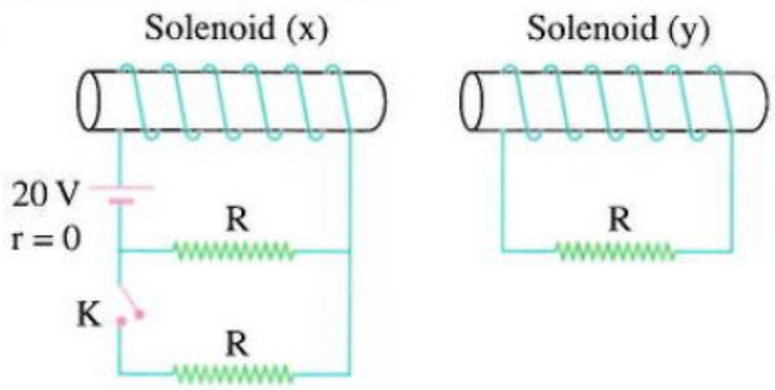
(a) equals ($\theta$) to the right side of the zero scale

(b) equals ($\theta$) to the left side of the zero scale

(c) greater than ($\theta$) to the right side of the zero scale

(d) greater than ($\theta$) to the left side of the zero scale

**21** * In the opposite figure, the solenoids (x), (y) are of negligible resistances and the mutual induction coefficient between them is 0.01 H. Solenoid (x) carries a DC current of constant intensity. When closing the switch (K), the value of the current changes till becomes constant through an interval of 0.25 s and during that interval the average induced emf in the solenoid (y) becomes 0.04 V, so value of the resistor (R) equals ..... .




(a) 10 Ω (b) 20 Ω (c) 40 Ω (d) 60 Ω

**22** Figure (1) represents two adjacent solenoids P, Q and figure (2) represents graphically the relation between the current intensity (I) in coil P and the time (t),

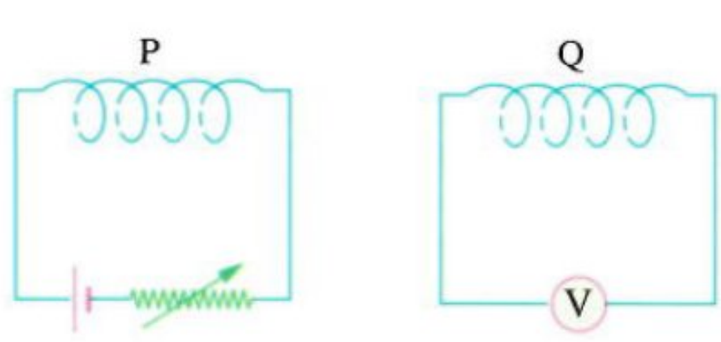



**Figure (1)**

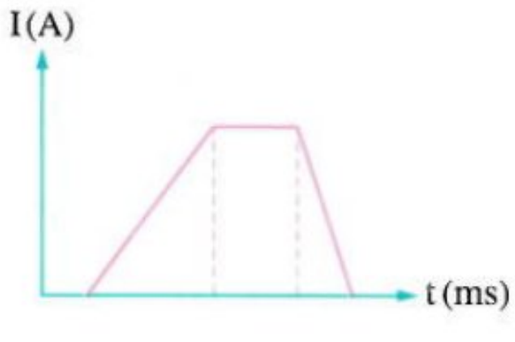
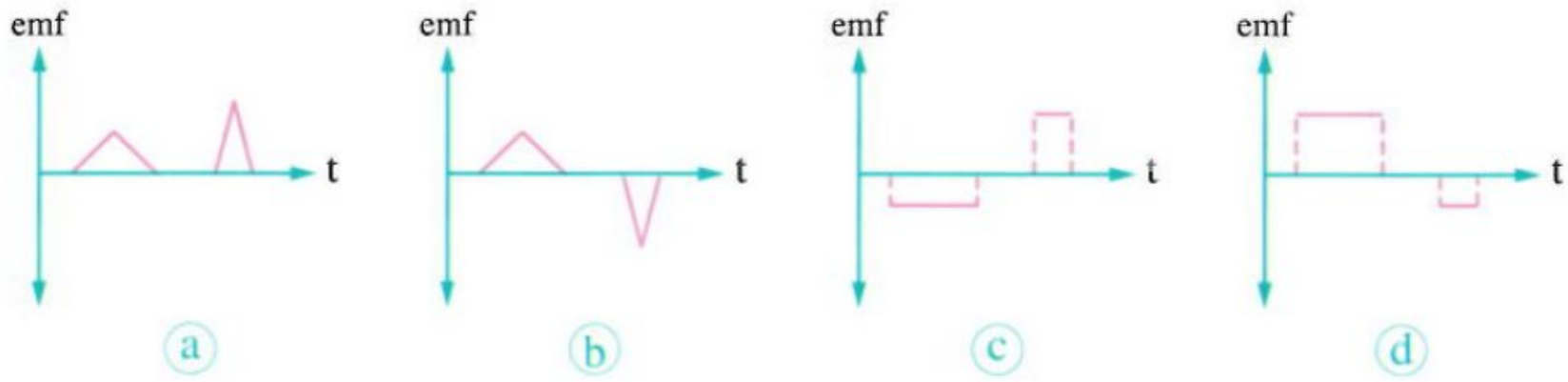


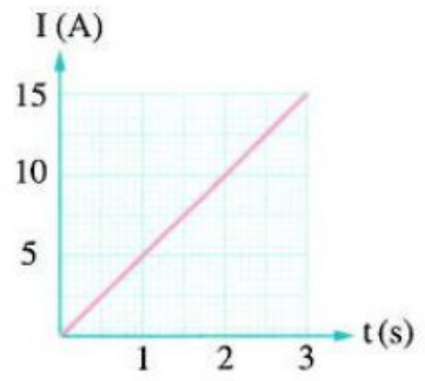

**Figure (2)**

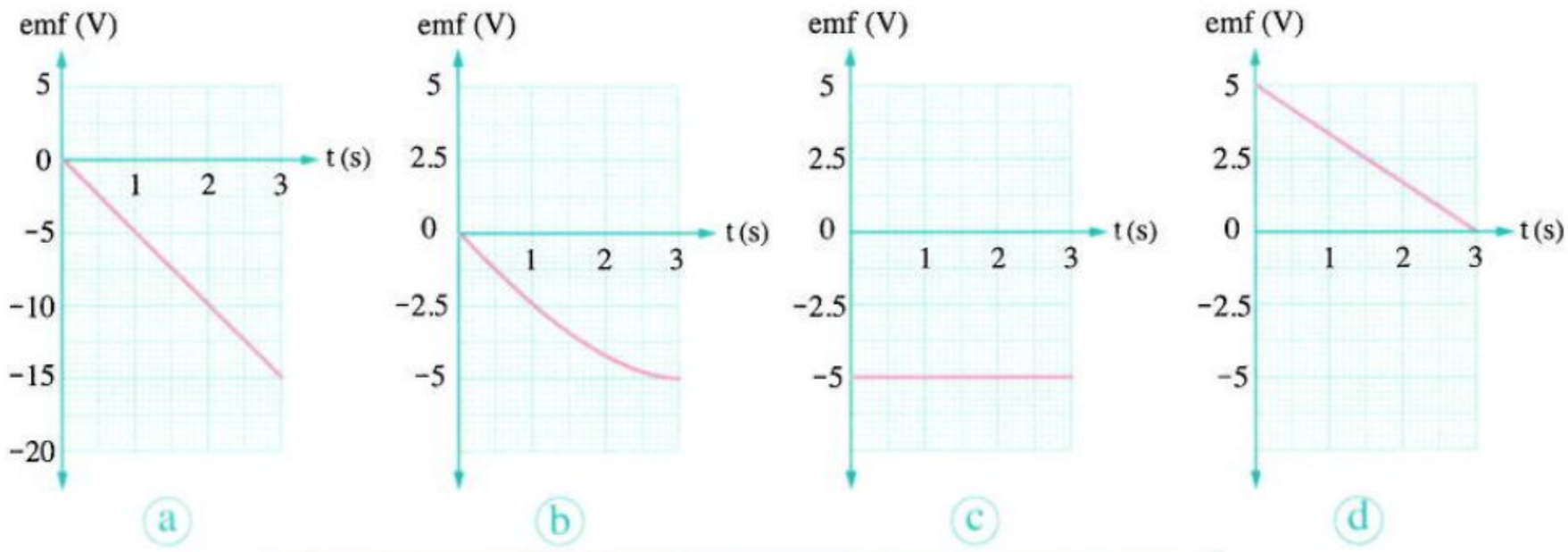
then the graph that represents the induced emf in coil Q versus time is ..... .



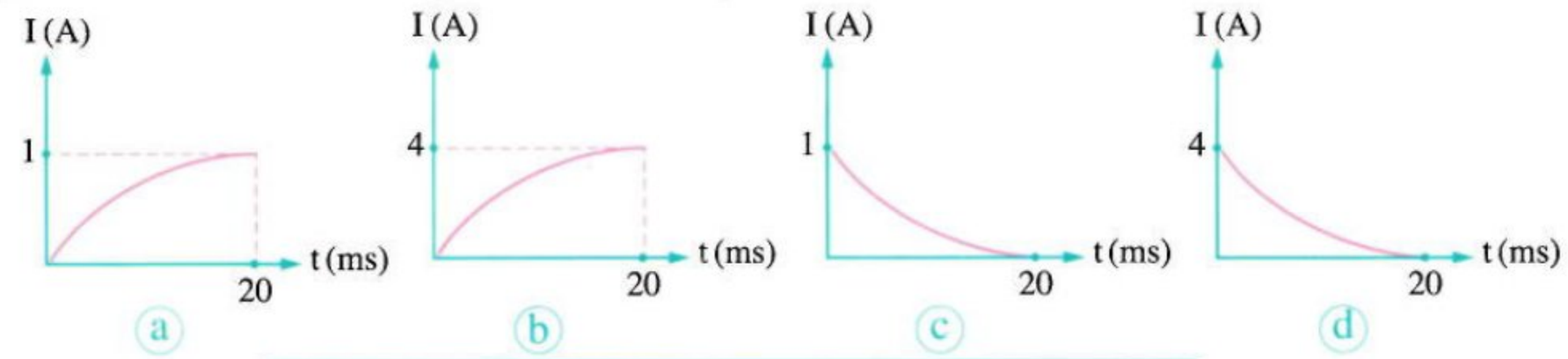

(a) (b) (c) (d)

23 The mutual induction coefficient between two adjacent coils is 1 H, if the passing current through one of the coils changes with time as in the opposite graph, then the graph that represents the relation between the induced emf generated in the second coil and the time is ..... .

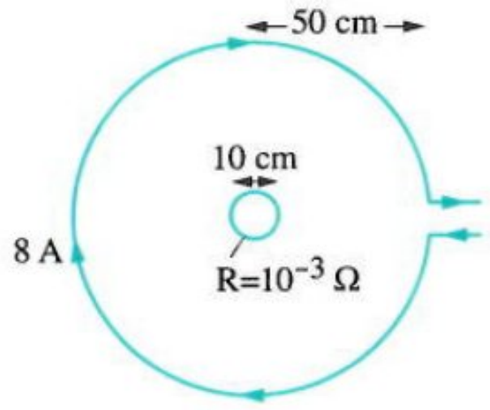



24 Two solenoids primary and secondary solenoid having mutual inductance between them of 10 mH. Which of the following graphs represents the change of the current of the primary solenoid (I) with time (t), so an average back induced emf of 2 V gets generated in the secondary solenoid through 20 ms?




25 The opposite figure shows a one turn small circular loop of radius 5 cm and resistance $10^{-3}$ Ω placed at the center of a one turn big loop of radius 50 cm. A variable current that changes uniformly from 0 to 8 A within $10^{-6}$ s passes through the big loop. Then the induced electric current passing in the small loop within the same period of time equals ..... .




(a) 49 A (b) 57 A (c) 66 A (d) 79 A

26 * Coil (y) of resistance 50 Ω, area 5 $cm^2$ that consists of 10 turns is placed in the same plane and at the center of a coil (x) of radius 11 cm that consists of 7 turns and carries an electric current of intensity I. During the rotation of coil x about an axis in its plane for 180° an electric charge of 20 nC has passed in coil y, so the current intensity I passing in coil x equals ..... .

(a) 2.5 A (b) 5 A (c) 7.5 A (d) 10 A



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27 ✳ A circular coil of diameter 2 cm that consists of 100 tightly adjacent turns is placed inside a long solenoid of diameter 5 cm that consists of 2100 turns/m such that both coils have the same axis. If the electric current intensity in the solenoid is decreased with a constant rate from 1.5 A to zero through an interval of 0.05 s, then increased to 1.5 A in the opposite direction at the same rate through an interval of 0.05 s, so the average emf induced in the circular coil equals ..... .

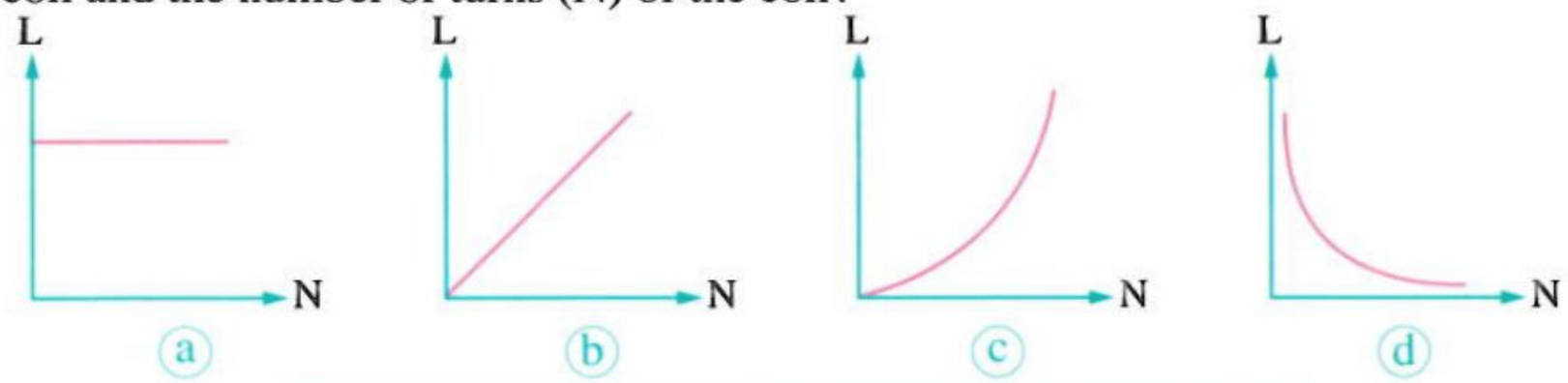
(a) $2.49 \times 10^{-3}$ V (b) $4.32 \times 10^{-3}$ V (c) $6.34 \times 10^{-3}$ V (d) $7.12 \times 10^{-3}$ V

## Self induction of the coil

28 The self inductance of a coil is measured in a unit called henry which is equivalent to ..... .

(a) volt.s (b) ohm.s (c) ohm/s (d) volt.s.ampere

29 Which of the following graphs represents the relation between the self inductance (L) of a coil and the number of turns (N) of the coil?




(a) (b) (c) (d)

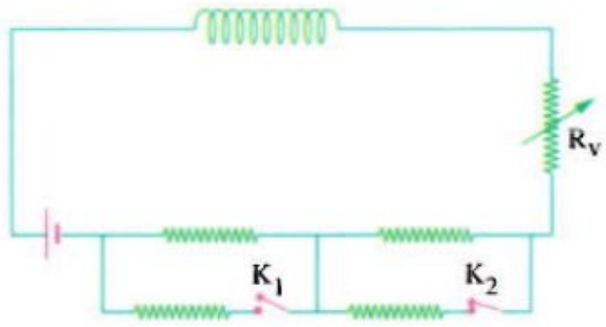
30 After a while of the flow of a direct electric current in an inductor, its intensity becomes constant due to ..... .

(a) generation of forward currents (b) generation of eddy currents
(c) vanishing of self induction (d) presence of back currents

31 The slow rate of growth of the current in a solenoid coil at the moment of closing its circuit is due to the ..... .

(a) generation of forward induced current
(b) generation of back induced emf
(c) loss of energy of the electric charges in the coil
(d) increase of the ohmic resistance in the circuit

32 In the opposite electric circuit, which of the following actions, each individually causes the generation of forward induced emf in the solenoid?

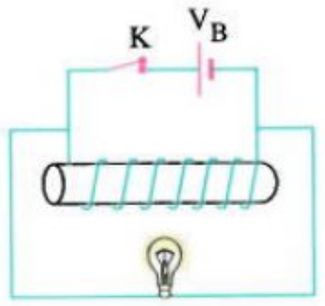



(a) Closing switch ($K_1$).
(b) Opening switch ($K_2$).
(c) Decreasing the value of the resistor ($R_v$).
(d) Inserting an iron rod in the solenoid.



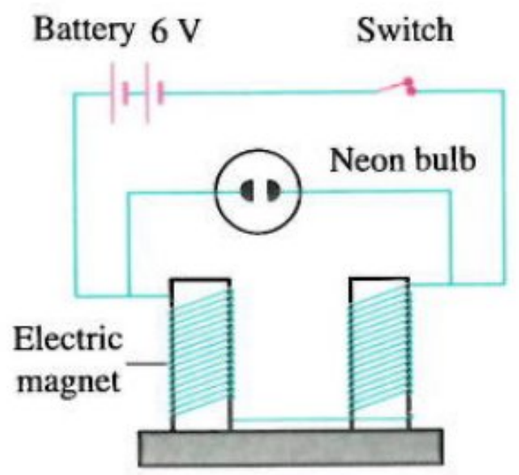
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**33** In the shown electric circuit, at opening the switch K, the brightness of the lamp ..... .




(a) increases gradually (b) decreases gradually
(c) increases instantly then vanishes (d) vanishes

**34** In the shown experiment, if the neon bulb works at a potential that is not less than 180 V, then:



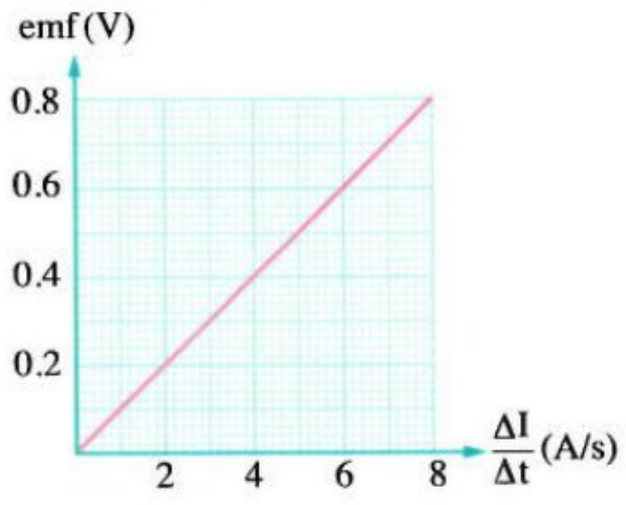

(i) The bulb lights at the moment of opening the switch due to ..... .

(a) the emf of the battery
(b) the back induced emf in the coil
(c) the forward induced emf in the coil
(d) no correct answer

(ii) The self-inductance of the coil is ..... .

(a) maximum at the moment of closing the switch
(b) minimum at the moment of opening the switch
(c) zero at the moment of opening the switch
(d) always constant

**35** The opposite graph represents the relation between the magnitude of the induced electromotive force in a coil (emf) due to self-induction and the rate of change in the electric current intensity which passes through it $\left(\frac{\Delta I}{\Delta t}\right)$, then:



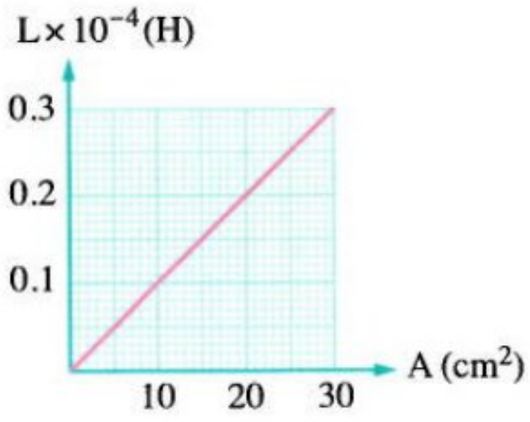

(i) The self induction coefficient of the coil (L) equals ..... .

(a) 0.1 H (b) 0.4 H (c) 0.6 H (d) 1 H

(ii) If the rate of change in current intensity in the coil is 5 A/s, then the induced electromotive force in the coil equals ..... .

(a) 0.45 V (b) 0.5 V (c) 0.55 V (d) 0.6 V

**36** The opposite graph represents the relation between the self-induction coefficient of a group of solenoids having the same length and their face areas. If the number of turns of each is 100 turns, the length of the solenoid is ..... .




(a) 0.1 π m (b) 0.2 π m
(c) 0.4 π m (d) 0.8 π m




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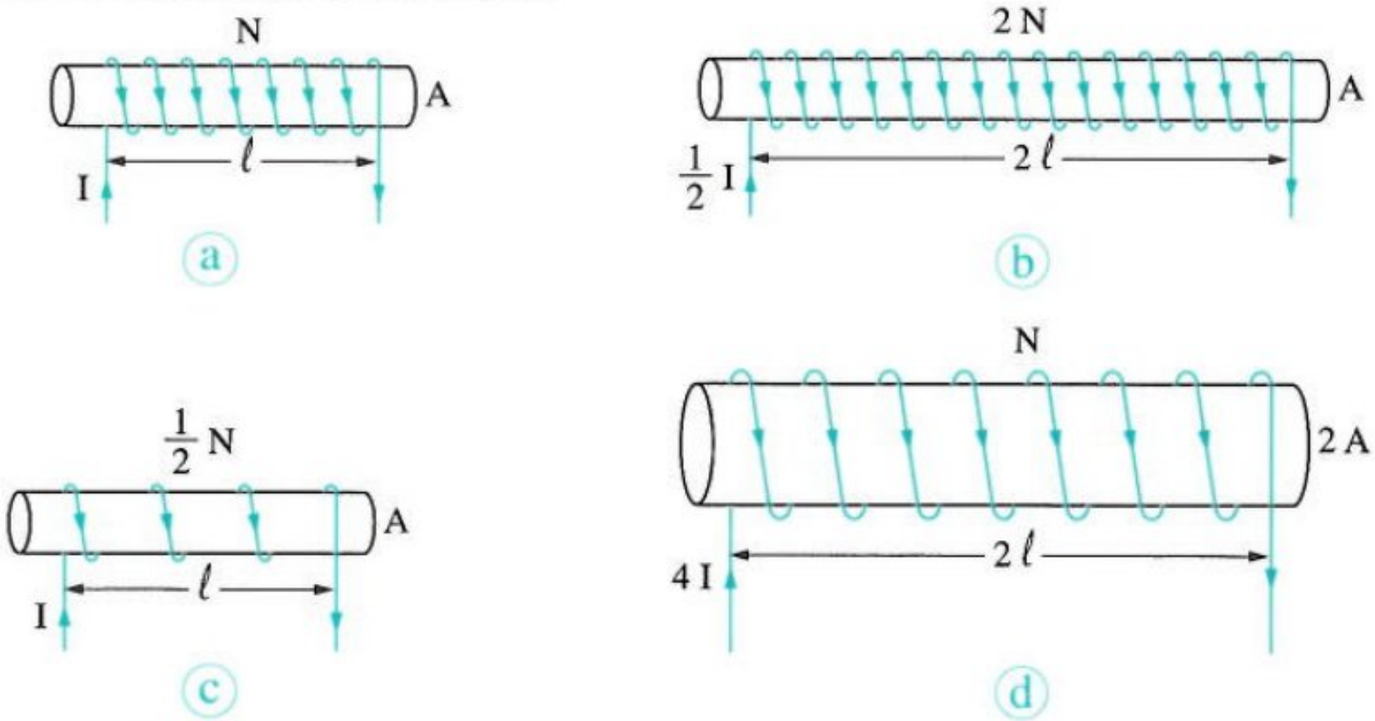
37 An inductor has a self-inductance L, so if each of the number of turns and the length of the inductor gets doubled holding the cross-sectional area constant, then its self-inductance becomes ..... .

(a) $\frac{L}{2}$ (b) L (c) 2 L (d) 4 L

38 Which of the following variations leads to increasing the self-inductance of a solenoid that is coiled on an iron rod to the double of its value when the other factors are kept constant?

(a) Increasing the number of turns to the double.
(b) Increasing the length of the solenoid to the double.
(c) Increasing the cross-sectional area of the solenoid to the double.
(d) Removing the iron rod from the solenoid.

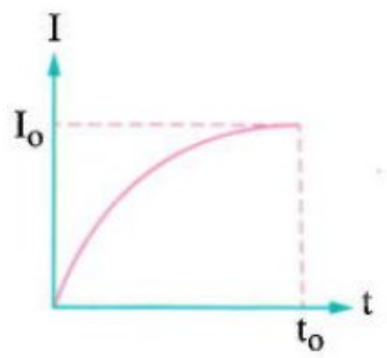
39 In which of the following cases, the self-induction coefficient of the coil is greater than the others if all coils have iron cores?




40 A coil of air core has a self inductance of 0.1 H, if an iron core is put inside it, its self induction coefficient will be ..... .

(a) equal to 0.1 H
(b) greater than 0.1 H
(c) less than 0.1 H
(d) depending on the intensity of the alternating current passing through it

41 The opposite graph represents the growth of the electric current in a hollow induction coil from the moment of closing its circuit till it reaches its final value ($I_o$), so if a core of soft iron is inserted inside the coil, the time interval taken by the current in the coil from the moment of closing its circuit to the moment of reaching its maximum value ($I_o$) is ..... .




(a) greater than $t_o$ (b) less than $t_o$
(c) equals $t_o$ (d) the answer is indeterminable



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42 A long inductor has a number of turns N and a self-inductance of 1 H. When an electric current of intensity 2 A passes through it, a magnetic flux of $4 \times 10^{-3}$ Wb is produced, so the number of turns of the inductor N equals ..... .

(a) 200 turns (b) 300 turns (c) 500 turns (d) 1000 turns

43 If the current intensity passing through a solenoid of 100 turns is changed by 4 A, hence the flux that is cut by the solenoid changes by 0.05 Wb during the same interval of time, so the self-inductance of the coil equals ..... .

(a) 4 H (b) 0.5 H (c) 1.25 H (d) 0.2 H

44 * A solenoid coil contains 300 turns and its self inductance is $6 \times 10^{-3}$ H, if the rate of change of its electric current intensity is 2 A/s. Then the rate of change in the magnetic flux produced through the coil equals ..... .

(a) $2 \times 10^{-5}$ Wb/s (b) $4 \times 10^{-5}$ Wb/s (c) $6 \times 10^{-5}$ Wb/s (d) $8 \times 10^{-5}$ Wb/s

45 * A coil is formed of 100 turns, its self induction coefficient is 0.03 H and it carries an electric current producing magnetic flux density of $6 \times 10^{-4}$ Wb. If the electric current passing through the coil is vanished within 0.02 s, then:

(i) The average induced emf generated in the coil equals ..... .

(a) 3 V (b) 6 V (c) 9 V (d) 12 V

(ii) The electric current intensity passing in the coil equals ..... .

(a) 0.5 A (b) 1 A (c) 2 A (d) 5 A

46 * Two solenoid coils, the first is of length $\ell$, face area A and number of turns N while the second is of length $\frac{1}{2}\ell$, face area 2 A and number of turns $\frac{1}{4}$ N. Then the ratio between their self induction coefficients $\left(\frac{L_1}{L_2}\right)$ equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{2}{1}$ (c) $\frac{4}{1}$ (d) $\frac{1}{4}$

47 A hollow solenoid has self-inductance of $2 \times 10^{-4}$ H and when it gets coiled around an iron core, its self-inductance becomes 0.3 H, then the ratio between the magnetic permeability of air and iron respectively is ..... .

(a) $\frac{1}{1600}$ (b) $\frac{1}{1000}$ (c) $\frac{1}{1500}$ (d) $\frac{3}{500}$

48 A straight wire of length $\ell$ has coiled in the form of a circular coil (x) of radius d and another time in the form of a circular coil (y) of radius 2d, so the ratio between the coefficients of the self-inductance of the two coils $\left(\frac{L_x}{L_y}\right)$ equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{2}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$



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49 ✳ A solenoid coil of 500 turns has length 0.4 m and the area of each turn is 40 $cm^2$, it was connected to an electric current source, thus electric current of 2 A passed through it. If the current is cut off within 0.1 s, then the forward induced emf formed in the coil is equal to ..... .
(Where: $\pi = 3.14$)

(a) $8.2 \times 10^{-2}$ V (b) $6.28 \times 10^{-2}$ V (c) $9.4 \times 10^{-3}$ V (d) $6.28 \times 10^{-3}$ V

50 ✳ A solenoid coil of 40 turns has length 10 cm and the area of each turn is $\frac{70}{22}$ $cm^2$, so:

(i) The self induction coefficient of the coil equals ..... .

(a) $3.2 \times 10^{-6}$ H (b) $4.8 \times 10^{-6}$ H (c) $6.4 \times 10^{-6}$ H (d) $7.6 \times 10^{-6}$ H

(ii) The value of the self induction coefficient of the coil if 10 turns are removed from it is equal to ..... .

(a) $1.5 \times 10^{-6}$ H (b) $2.2 \times 10^{-6}$ H (c) $3.4 \times 10^{-6}$ H (d) $4.8 \times 10^{-6}$ H

51 When the intensity of the electric current that is passing through an inductor changes in the rate of 20 A/s. An electromotive force of 5 V is induced in the coil, so the self induction coefficient of the coil is ..... .

(a) 0.18 H (b) 0.2 H (c) 0.22 H (d) 0.25 H

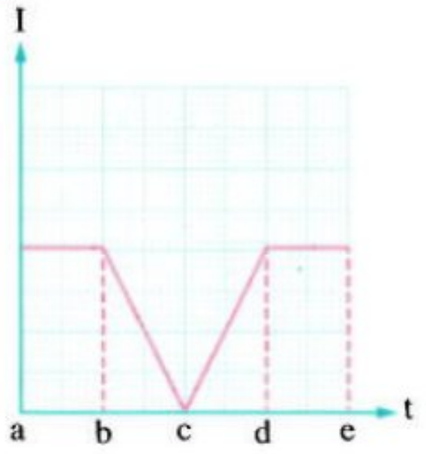
52 ✳ When the electric current intensity passing through a coil of self inductance 0.005 H changes from 10 A to zero, an induced emf of 5 V is generated between its terminals. Then the time of the change in the electric current equals ..... .

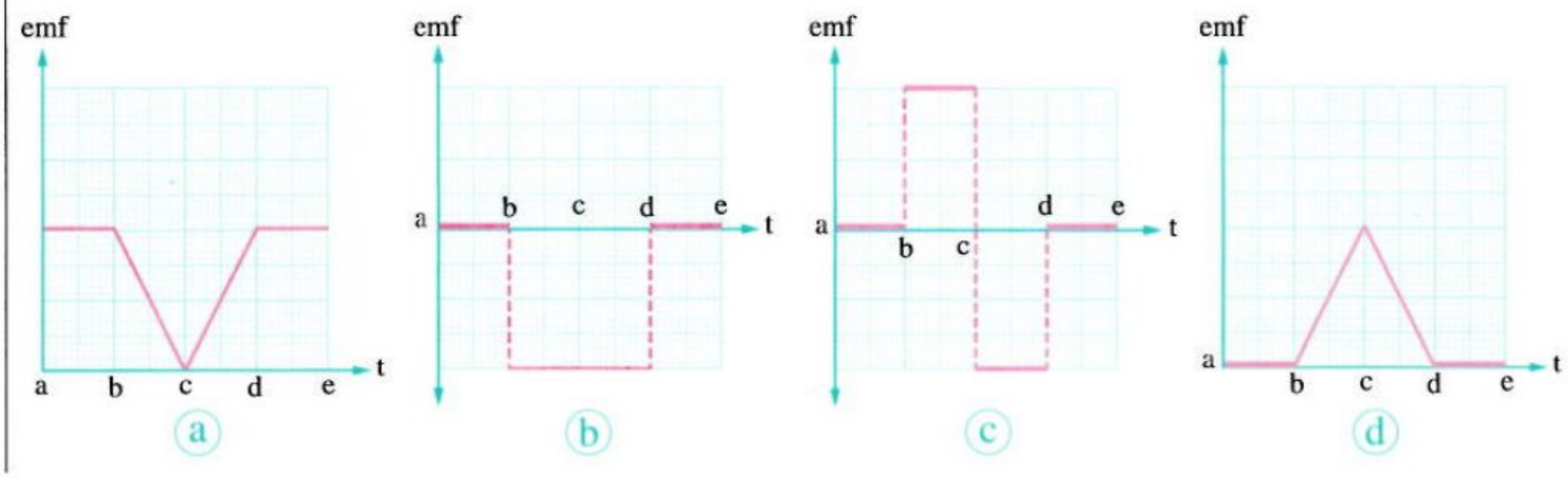
(a) 0.01 s (b) 0.02 s (c) 0.03 s (d) 0.04 s

53 Ohmic resistors are made of double folded coiled wires ..... .

(a) to decrease the resistance of the wire (b) to increase the resistance of the wire
(c) to avoid self-induction (d) to eliminate the resistance of the wire

54 ✳ The opposite graph shows the relation between the electric current intensity (I) and the time (t) in an inductor. Which of the following graphs represents the relation between the induced emf in the inductor and the time (t)?

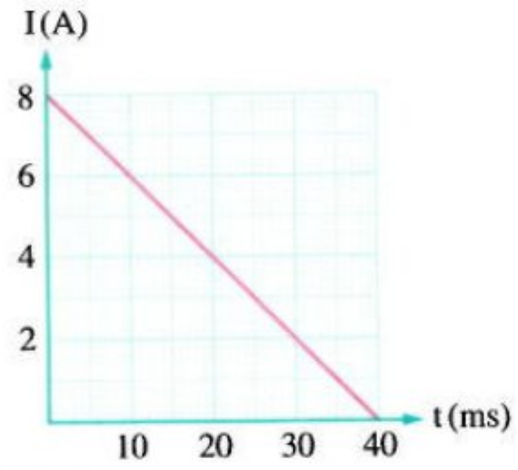




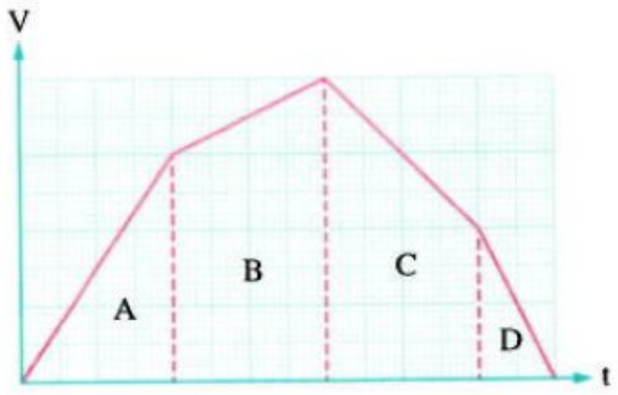

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**55** The opposite figure represents the graphical relation between the electric current intensity (I) and time (t) in a solenoid. If the self-inductance of the coil is 60 mH, then the induced electromotive force in it equals ..... .



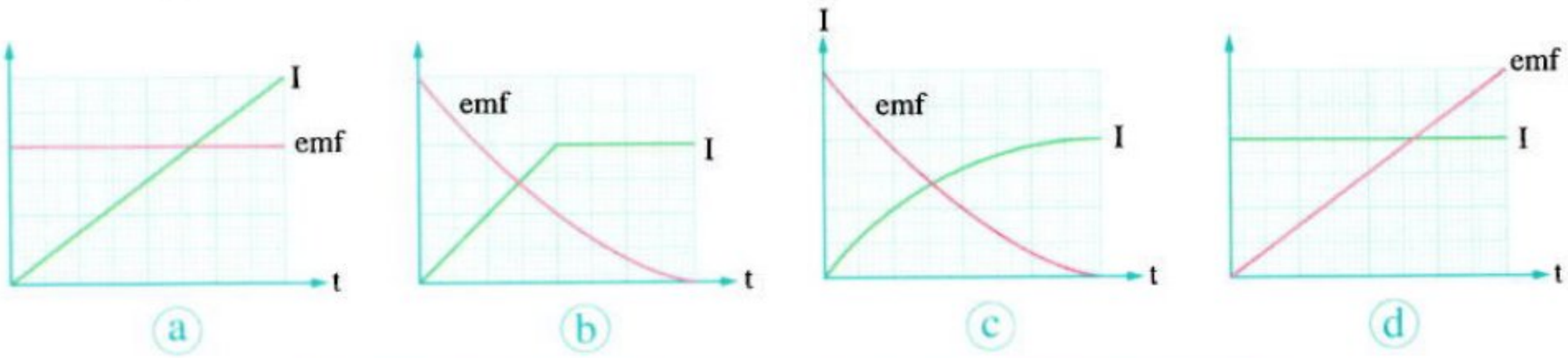

(a) 10 V (b) 12 V
(c) 20 V (d) 24 V

**56** An induction solenoid of resistance (R) in a closed electric circuit where the opposite graph represents the change of the voltage (V) across its ends with time (t). So, the greatest forward induced emf gets generated in the solenoid during the interval ..... .




(a) A (b) B
(c) C (d) D

**57** An electric circuit consists of an inductor of ohmic resistance R, battery and a switch all are connected in series where the graph that represents the change in each of the induced emf in the solenoid and the value of the current passing through it with time (t) at closing the circuit is ..... .




**58** In the shown electric circuit in figure (1), when each of the two switches ($K_1$), ($K_2$) is closed individually, the electric current passing in the circuit grows as represented graphically by the curves ①, ② respectively in figure (2), so:

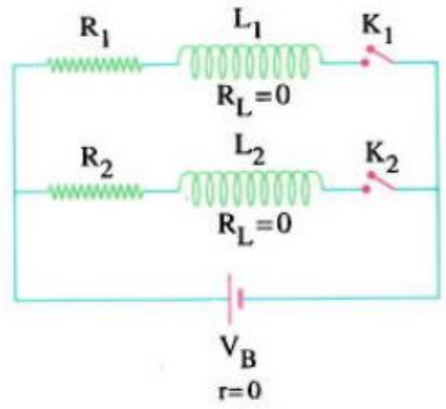



Figure (1)

(i) The relation between the self-induction coefficient for the two solenoids ($L_1$), ($L_2$) is ..... .

(a) $L_1 > L_2$ (b) $L_1 < L_2$
(c) $L_1 = L_2$ (d) the answer is indeterminable

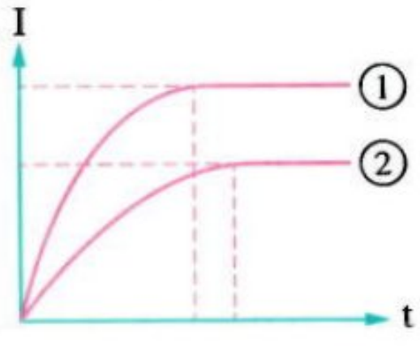



Figure (2)

(ii) The relation between the values of the resistors ($R_1$), ($R_2$) is ..... .

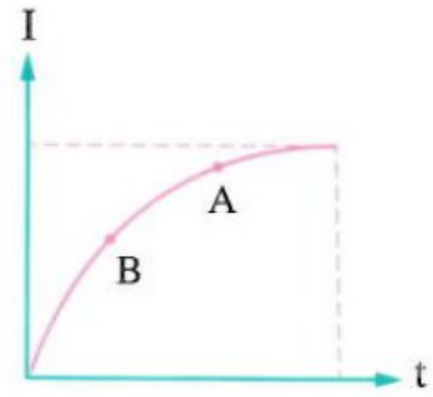
(a) $R_1 > R_2$ (b) $R_1 < R_2$
(c) $R_1 = R_2$ (d) the answer is indeterminable



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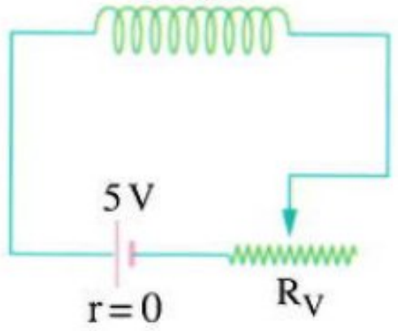
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59 The opposite graph represents the growth of the current through induction coil that is connected to a battery from the moment of closing its circuit, so the physical quantity which its value at point B is less than that at point A is ..... .




(a) self-inductance of the coil
(b) back induced emf in the coil
(c) the time rate of change in the electric current
(d) the net emf in the circuit

60 The opposite figure shows an electric circuit which contains a battery of emf 5 V with a negligible internal resistance, an inductor of inductance 10 mH with a negligible ohmic resistance and a variable resistance ($R_v$). If the variable resistance ($R_v$) is increased gradually to have the value of 10 Ω at a certain instant, the electric current passing in the circuit at that instant will be ..... .




(a) equal to 0.5 A
(b) greater than 0.5 A
(c) less than 0.5 A and doesn't equal zero
(d) equal to zero

61 * A coil of resistance 15 Ω and self induction coefficient 0.6 H is connected to DC source of 120 V, so the rate of increase of the electric current in the following cases:

(i) At the moment of connection equals ..... .

(a) 50 A/s
(b) 100 A/s
(c) 175 A/s
(d) 200 A/s

(ii) At the moment of reaching 80% of the maximum value of the electric current equals ..... .

(a) 20 A/s
(b) 40 A/s
(c) 60 A/s
(d) 80 A/s

62 * Two adjacent coils A, B consist of 500, 2000 turns respectively. When the electric current in coil A changes by 10 A, the magnetic flux in coil A changes by $2 \times 10^{-3}$ Wb and in coil B changes by $10^{-4}$ Wb, then:

(i) The coefficient of self induction of coil A equals ..... .

(a) 0.1 H
(b) 0.4 H
(c) 0.6 H
(d) 0.8 H

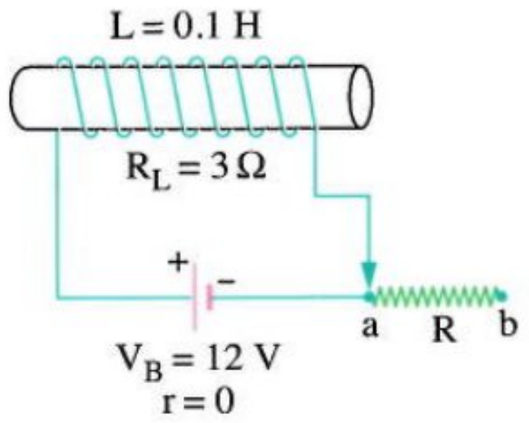
(ii) The mutual induction coefficient between the two coils equals ..... .

(a) 0.08 H
(b) 0.05 H
(c) 0.02 H
(d) 0.01 H



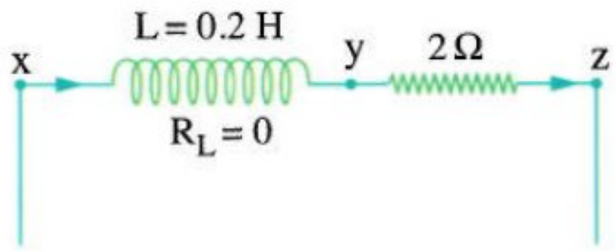
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63 ✳ An electric circuit consists of a battery of emf 12 V and negligible internal resistance, an induction solenoid of self-induction 0.1 H of ohmic resistance 3 Ω and a rheostat (R) as in the opposite figure. When the slider is moved from position (a) to position (b) through a time interval of 0.02 s, an induced emf of 15 V is generated in the solenoid, so value of the rheostat resistance R when the slider is at position (b) equals ..... .



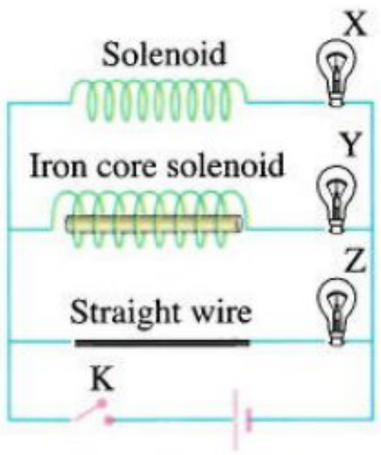

(a) 3 Ω (b) 6 Ω (c) 9 Ω (d) 12 Ω

64 The opposite figure represents a part of an electric circuit which contains a resistor of resistance 2 Ω and induction solenoid of self-inductance 0.2 H with negligible ohmic resistance. If the value of the current passing through the solenoid is increasing in the rate of 20 A/s at a certain moment while value of the current at that moment was 2 A, so the potential difference between the points x, z equals ..... .




(a) 0 (b) 4 V (c) 8 V (d) 12 V

65 ✳ A straight wire and two solenoids of equal ohmic resistances are connected to three identical lightbulbs (x, y, z), a battery and a switch as shown in the opposite figure, so the correct order of bulbs according to the required time interval to reach their maximum brightness is ..... .




(a) z = y = x (b) z > x > y (c) y > x > z (d) y = x > z

66 ✳ Two insulated overlapped solenoids (x, y) are equal in length and cross-sectional area, their number of turns are $N_x$, $N_y$ respectively. They are coiled on a wrought iron core. Coil (x) is connected to an electric cell and a rheostat and its self-inductance is ($L_x$), while coil (y) is connected to a galvanometer and coiled around coil x. So, the mutual inductance (M) between the two coils equals ..... .

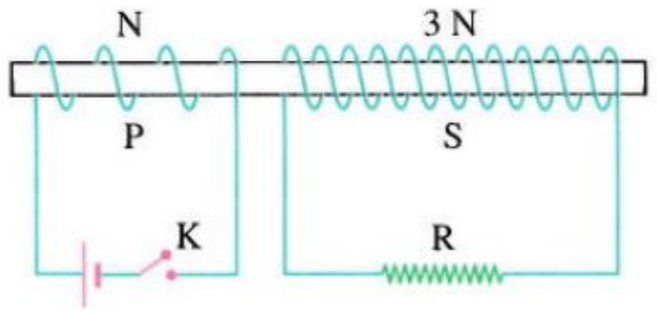
(a) $L_x \frac{N_x}{N_y}$ (b) $L_x \frac{N_y}{N_x}$ (c) $L_x \left(\frac{N_x}{N_y}\right)^2$ (d) $L_x \left(\frac{N_y}{N_x}\right)^2$



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67 In the opposite figure, two adjacent solenoids S, P of number of turns N, 3 N respectively, when closing the circuit of coil P, the change in the passing magnetic flux through the coil S is 60% of the change in the magnetic flux passing through the coil P, so the ratio between the mutual inductance between the coils and the self-inductance of coil P respectively equals ..... .




(a) 0.2 (b) 0.6 (c) 1.2 (d) 1.8

68 ✱ Two adjacent coils are coiled around a soft iron rod. The terminals of the primary coil are connected to a battery of emf of 20 V and a switch in series, so at the moment of closing the circuit of the primary coil, an induced emf of magnitude 5 V is generated across the terminals of the secondary coil. If you know that the self induction coefficient of the primary coil is 0.04 H, then the mutual induction coefficient between the two coils equals ..... .

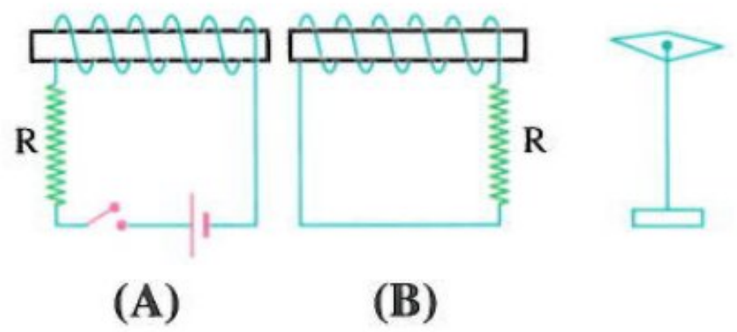
(a) $10^{-3}$ H (b) 0.01 H (c) 0.05 H (d) 0.1 H

## Second Essay questions

### The mutual induction between two coils

1 In the opposite figure:

**What** is the type of the magnetic pole of the magnetic needle that approaches coil B in the following cases:




(a) When closing the circuit of coil A?

(b) When approaching coil A after closing its circuit towards coil B?

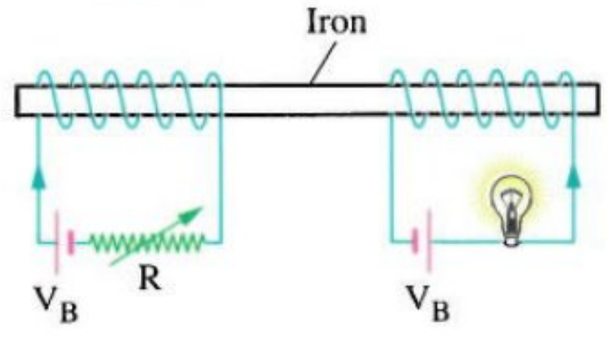
(c) When moving coil A after closing its circuit away from coil B?

(d) When opening the circuit of coil A?

2 In the opposite figure:

**What happens** to the brightness of the lamp when you increase the resistance R uniformly?

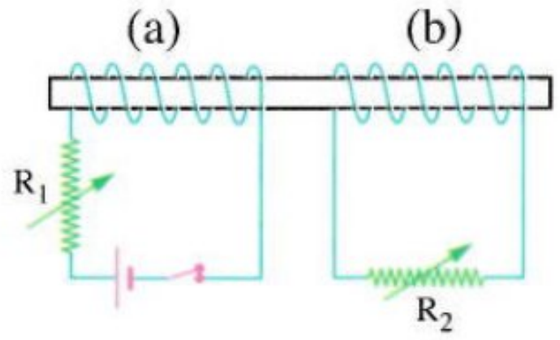
**Give reasons.**






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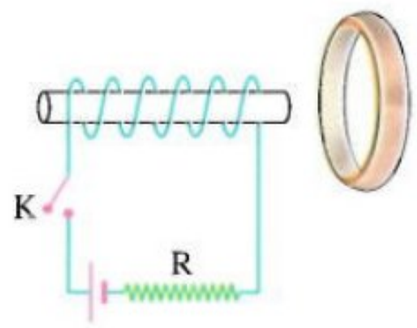
3 The opposite figure shows two electric circuits containing variable resistors $R_1$, $R_2$, **so in which of the following cases an emf gets generated in coil b? Give reasons.**




**(a)** During increasing $R_1$

**(b)** During increasing $R_2$

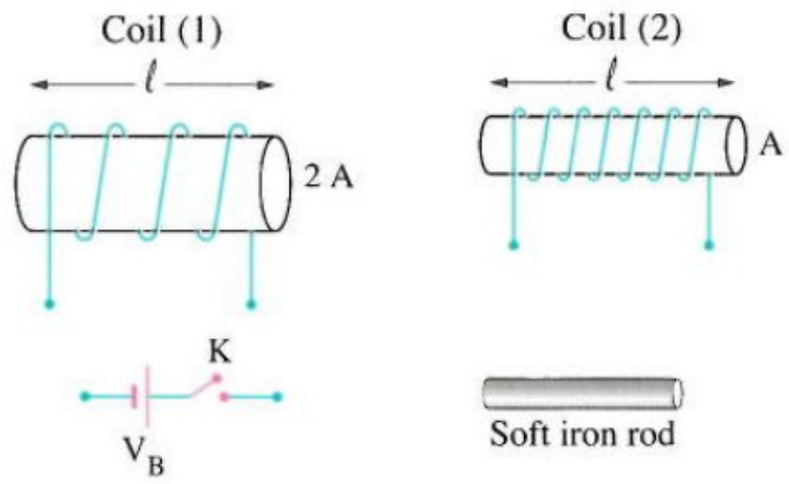
4 The opposite figure shows a metal ring placed in front of one of the faces of a solenoid such that the plane of the ring is perpendicular to the axis of the solenoid, **determine the direction of the induced current (if there is any) in the face of the ring that faces the coil:**




**(a)** At the moment of closing switch K.

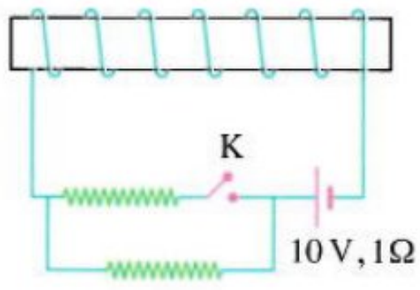
**(b)** After the current reaches its constant value (according to Ohm's law) in the solenoid circuit.

5 The opposite figure shows two solenoids (1), (2), battery $V_B$, switch (K) and a soft iron rod, **describe** how those components can be configured so that the mutual inductance between the two coils becomes as high as possible.

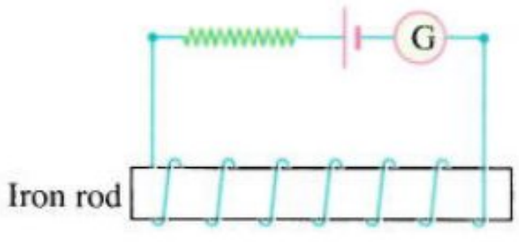



## Self induction of the coil

6 In the opposite circuit, **will** the induced emf in the coil be back or forward at the moment of closing switch K? **Explain your answer.**




7 In the opposite circuit, **describe with explanation what happens to** the galvanometer reading at pulling out the iron rod from the coil.





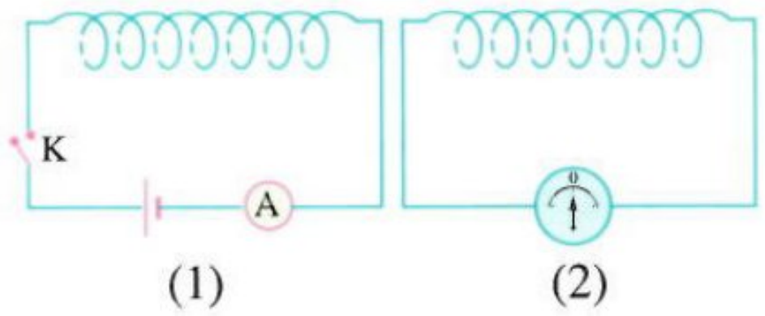

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8 In the opposite figure:

Coil (1) is connected in series with an electric cell, switch (K) and an ammeter (A), while coil (2) is connected to a sensitive galvanometer having its zero at the middle of the scale.




**Mention giving reasons what do you observe on the reading of the ammeter and the galvanometer in the following two cases:**

(a) At the moment of closing switch (K).

(b) At the moment of closing switch (K) while having soft iron cores inside both coils.

9 A solenoid of length 1.1 m and cross-sectional area 10 $cm^2$ that consists of 700 turns carries an electric current of 2 A, **calculate:**

(a) The induced emf in the coil if the current has vanished through 0.01 s.

(b) The self inductance of the coil.

10 When increasing the self inductance of an inductor, **explain what happens to** the time of each of:

(a) The current growth at the moment of closing the coil circuit.

(b) The current decay at the moment of opening the coil circuit.

11 **Mention the physical quantities measured in the following units and its equivalent unit:**

(a) Ω.s

(b) $T.m^2/s$

(c) V.s

(d) Ω.C

(e) V.s/A.m

(f) $V.s/m^2$

(g) J.s/A.C

(h) Wb/A

12 **Why** are the wires of standard resistors coiled double folded?



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# Questions of the Previous Years Exams ? on Lesson 2

## The mutual induction between two coils

**1** The graph represents the relation between the induced electromotive force (emf) in the secondary coil and the rate of change of the electric current intensity in the adjacent primary coil $\left(\frac{\Delta I}{\Delta t}\right)$, so the mutual inductance between the two coils equals ..... . *(Exp. June 21)*



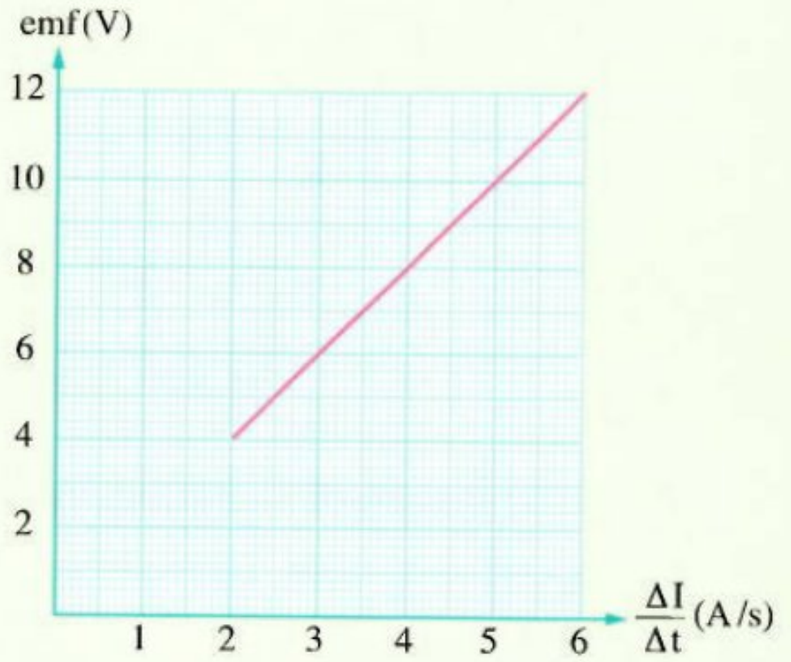

(a) 0.05 mH

(b) 50 mH

(c) 0.04 mH

(d) 40 mH

**2** The opposite graph represents the relation between the induced electromotive force (emf) in the secondary coil and the rate of change of the electric current intensity in the adjacent primary coil $\left(\frac{\Delta I}{\Delta t}\right)$. So, the mutual inductance between the two coils is ..... . *(1st session 21)*




(a) 0.5 H

(b) 1.6 H

(c) 2 H

(d) 6 H



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**3** The graph (1) represents the relation between the induced electromotive force (emf) in the secondary coil and the rate of change of the electric current intensity in the adjacent primary coil $\left(\frac{\Delta I}{\Delta t}\right)$.

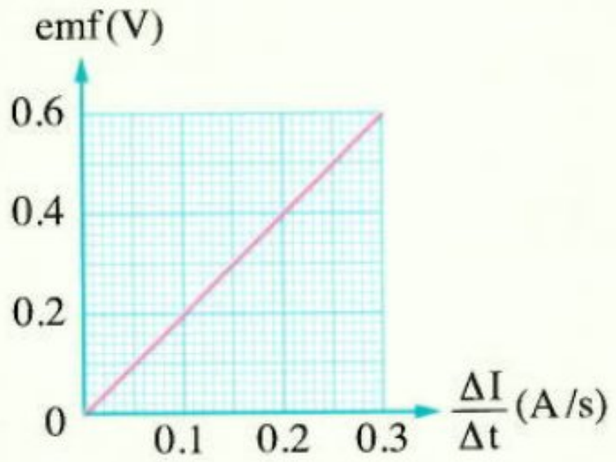



**Figure (1)**

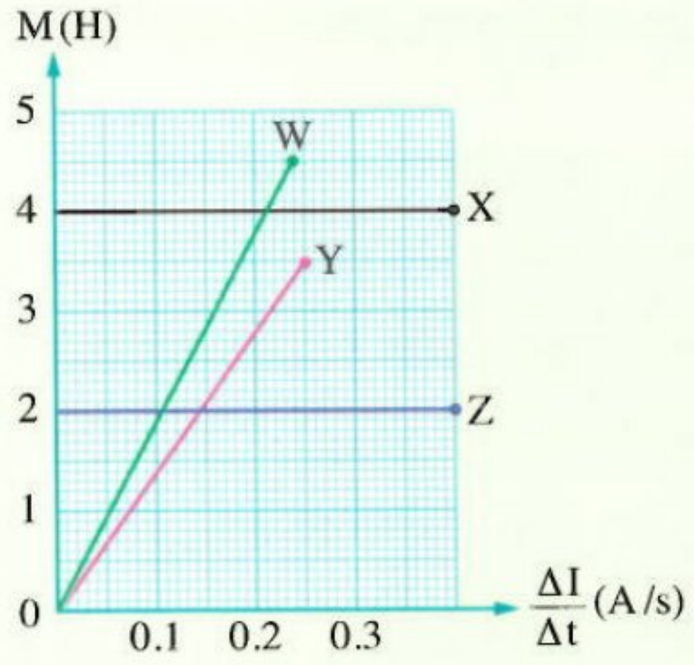
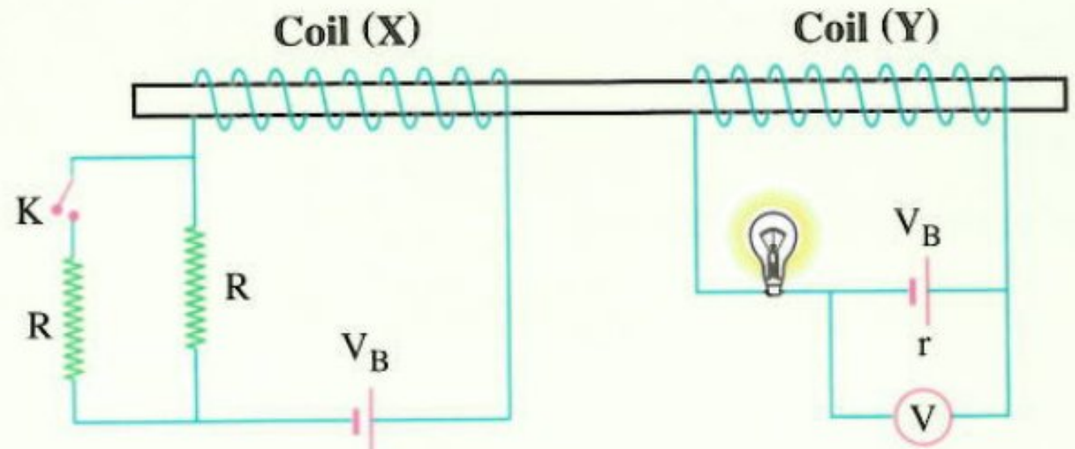



**Figure (2)**

Which of the graphical lines (W, X, Y, Z) in graph (2) represents the relation between mutual inductance between those two coils (M) and the rate of current change in the primary coil? *($2^{nd}$ session 21)*

(a) W (b) X (c) Y (d) Z

**4** The figure illustrates two adjacent similar coils (X), (Y).




So, at the instant of switching on the key (K) in the circuit of coil (X), ..... . *($1^{st}$ session 22)*

(a) the illumination of the bulb decreases while the voltmeter reading increases

(b) the illumination of the bulb increases while the voltmeter reading decreases

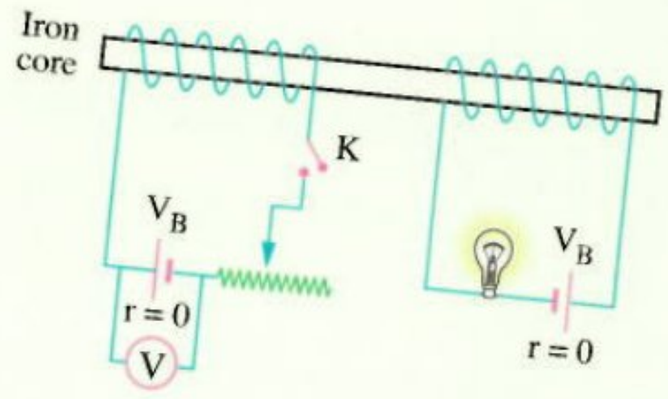
(c) both the illumination of the bulb and the voltmeter reading decreases

(d) both the illumination of the bulb and the voltmeter reading increases



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**5** Two adjacent coils with the same iron core as in the figure. At the instant of switching on the key (K) ..... .




(a) the illumination of the bulb increases and the voltmeter reading remains constant

(b) the illumination of the bulb decreases and the voltmeter reading increases

(c) the illumination of the bulb decreases and the voltmeter reading decreases

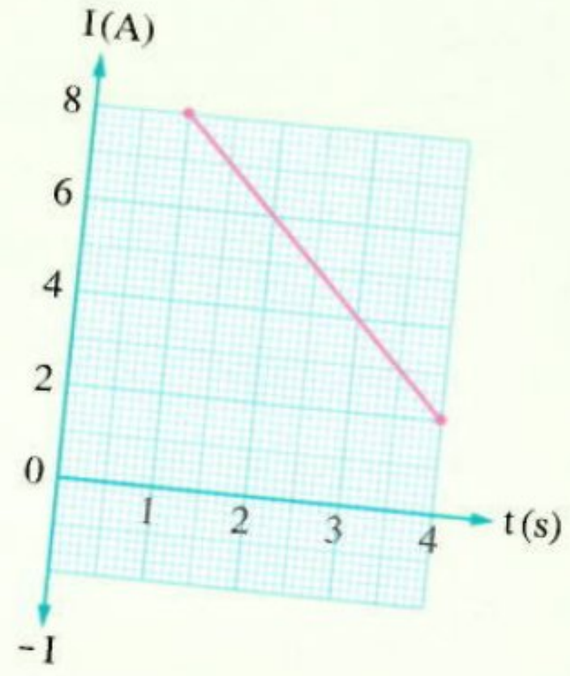
(d) the illumination of the bulb decreases and the voltmeter reading remains constant

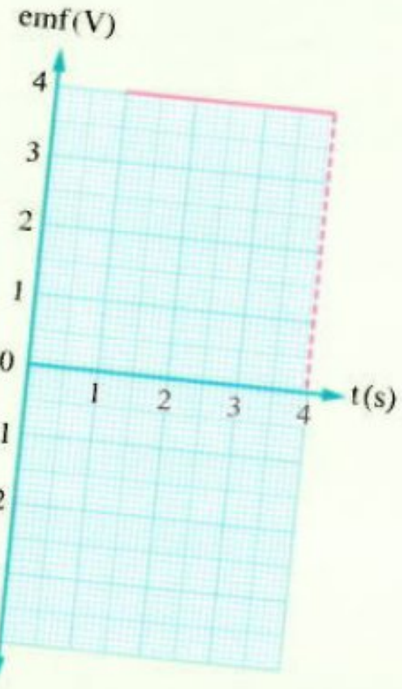
*(2nd session 22)*

**6** Two adjacent coils the mutual inductance between them is (2 H). The graph represents the relation between the change of the electric current (I) in the primary coil with time (t).

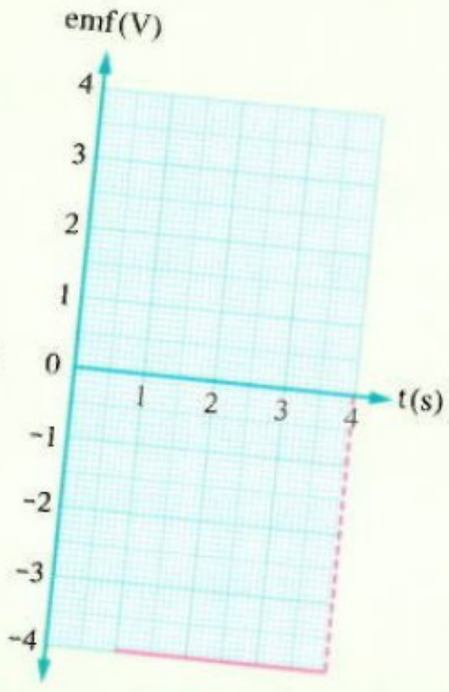
Which one from the following graphs represents the relation between the induced electromotive force (emf) in the secondary coil with time (t)?

*(1st session 23)*

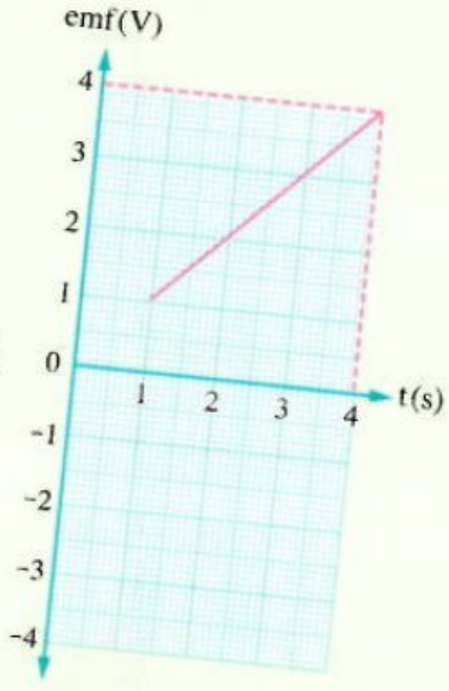


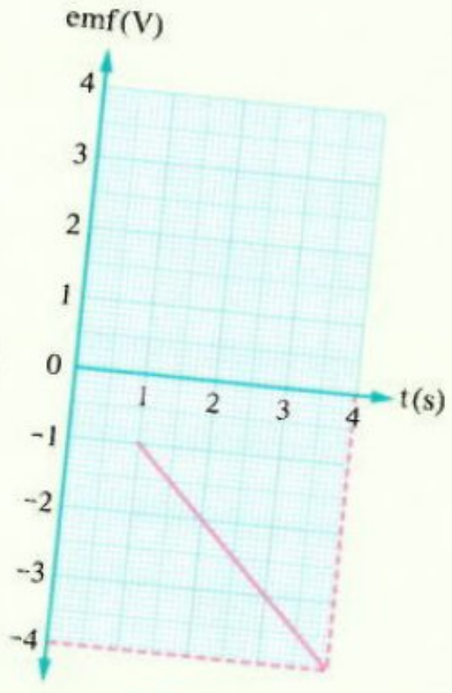

(a)




(b)




(c)




(d)

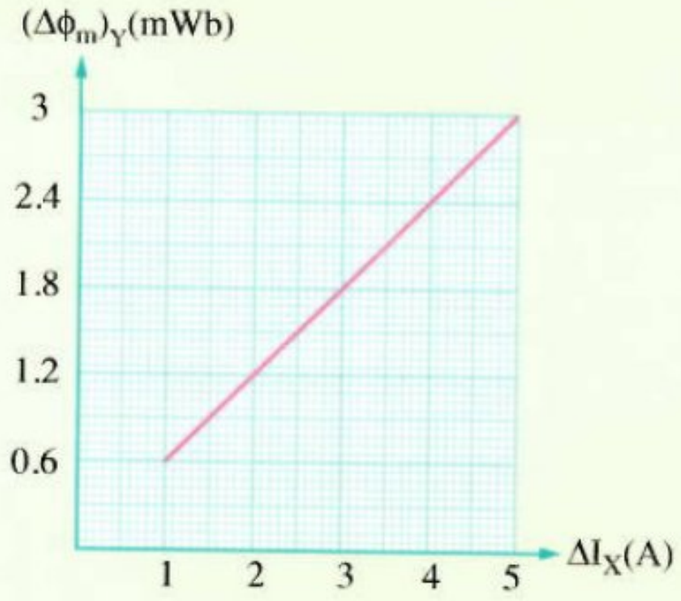


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**7** Two coils (X), (Y), the number of turns of coil (X) is 500 turns and the number of turns of coil (Y) is 1000 turns. The graph represents the change of magnetic flux $(\Delta\phi_m)_Y$ through coil (Y) when the current $(I_x)$ changes through coil (X). So, the mutual inductance between the two coils equals ..... .

*(2nd session 23)*




(a) 0.3 H (b) 0.6 H (c) 0.9 H (d) 1.2 H

**8** Two solenoids one inside the other, when the electric current intensity in one of them changes from 4 A to zero through 0.01 s, an induced emf of 40 V gets generated in the other solenoid, so the mutual induction coefficient between the two solenoids equals ..... .

*(Practice 25)*

(a) 0.1 H (b) 0.15 H (c) 0.2 H (d) 0.25 H

**9** The unit (henry.$m^{-1}$) is the measuring unit of ..... .

*(Practice 25)*

(a) the magnetic permeability of the medium ($\mu$)

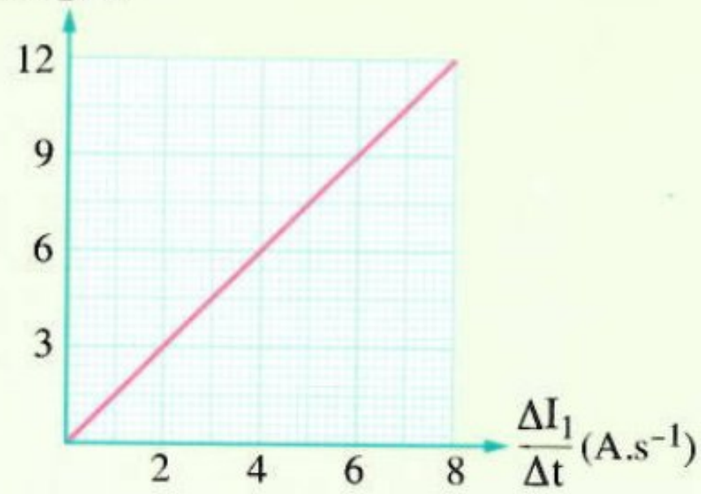
(b) magnetic dipole moment ($m_d$)

(c) magnetic flux density (B)

(d) the magnetic flux ($\phi_m$)

**10** Two adjacent coils (X and Y), the graph represents the relationship between the induced electromotive force ($emf_2$) in coil (Y) and the rate of change of current $(\frac{\Delta I_1}{\Delta t})$ passing through coil (X). What is the coefficient of mutual induction between the two coils?

*(1st session 25)*




(a) 0.5 H (b) 3 H

(c) 1.5 H (d) 0.25 H



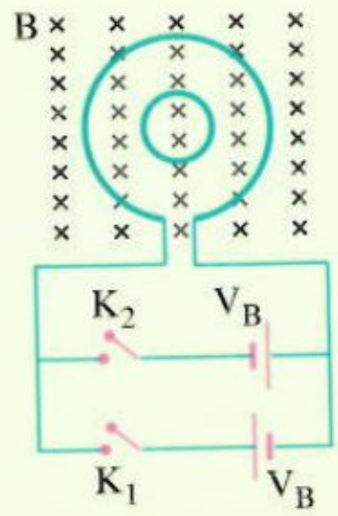
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**11** There are two adjacent coils, $L_1$ and $L_2$. An alternating current of frequency 50 Hz flows through coil $L_1$, and its value changes from zero to 2 A over the period of a quarter cycle of current. If the mutual inductance between the two coils is 1.5 H, the average induced emf in coil $L_2$ over this period of time is equal to ..... .

*(2nd session 25)*

(a) 600 V
(b) 500 V
(c) 400 V
(d) 300 V

**12** Two concentric metal rings at the same plane are exposed to a uniform magnetic field with flux density B that its direction is perpendicular into the page. The large ring is connected to two batteries each of negligible internal resistance and having the same electromotive force ($V_B$) and open switches $K_1$ and $K_2$, as shown in the figure. When only $K_1$ is closed, the larger ring produces a magnetic field with a flux density of 2B, so an average induced electromotive force is generated in the small ring of value (emf). When only $K_2$ is closed while $K_1$ is opened, an average induced electromotive force gets generated in the small ring of value ..... .




(Assuming that the magnetic field produced by the large ring is uniform over the entire area of the small ring)

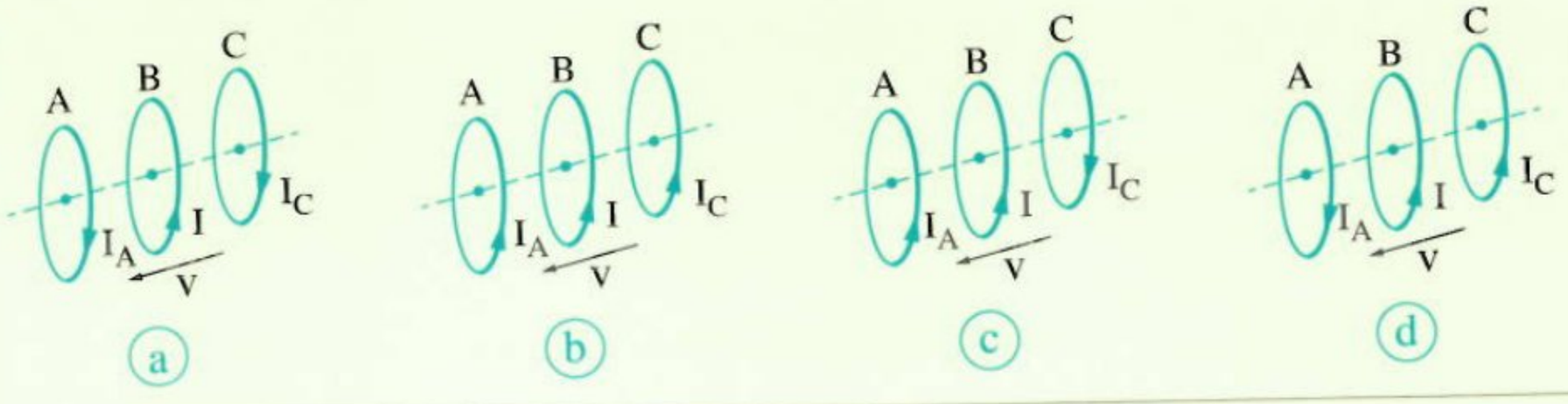
*(Practice 26)*

(a) 0.5 emf
(b) emf
(c) 2 emf
(d) 3 emf

**13** There conducting rings A, B, C, ring B carries a current (I) and moves at speed v as shown in the following figures.

So, the figure that shows the correct direction of the induced current produced in the other rings (A, C) is ..... .

*(Practice 26)*






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**14** A circular coil of diameter 0.02 m and number of turns 100 turns is placed at the center of a long solenoid of diameter 0.1 m and number of turns per unit length 2000 turn/m such that they have the same axis. If the intensity of the current passing in the solenoid is decreased with a constant rate from 4 A to zero within a time interval of 0.5 s, **calculate** the quantity of the total induced electric charge in the circular coil which passes in cross-section of its wire during that interval. *(Practice 25)*

(Where: resistance of the circular coil wire = $10\ \pi^2\ \Omega$, the magnetic permeability of the medium = $4\ \pi \times 10^{-7}$ Wb/A.m)

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## Self induction of the coil

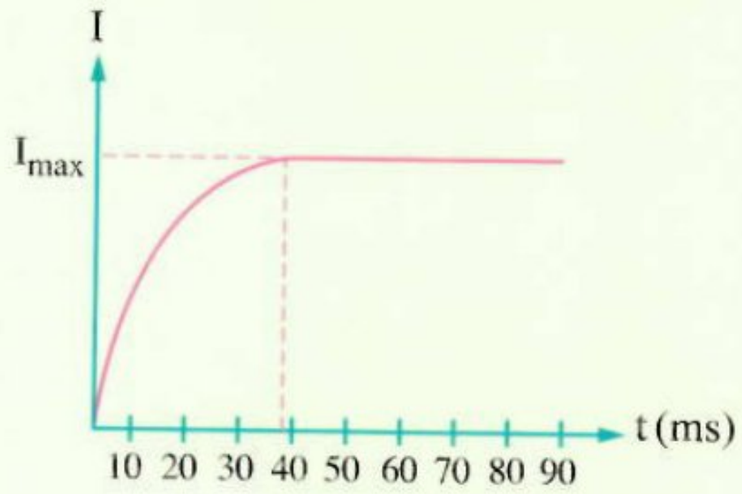
**15** Graph no (1) represents the growth of electric current in an inductor of inductance L that is connected to a battery at the instant of switching on the circuit. Which of the curves in figure (2) represents the growth of current if a soft iron core exists inside the coil at the moment of switching on the circuit? *(Exp. May 21)*

(a) Curve (1).

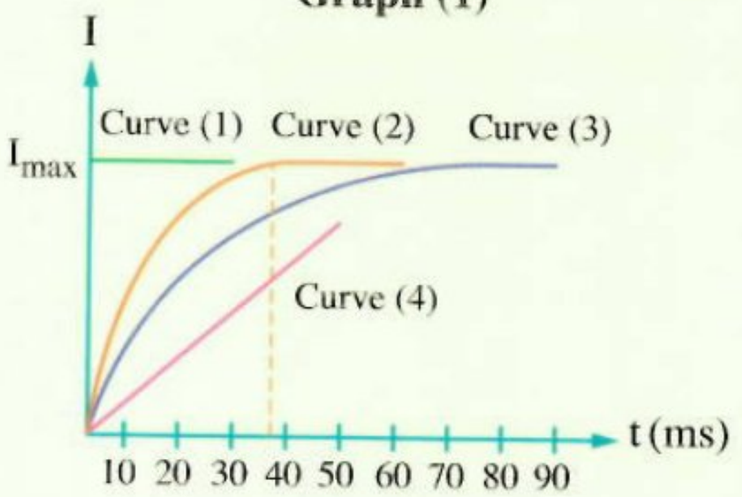
(b) Curve (2).

(c) Curve (3).

(d) Curve (4).

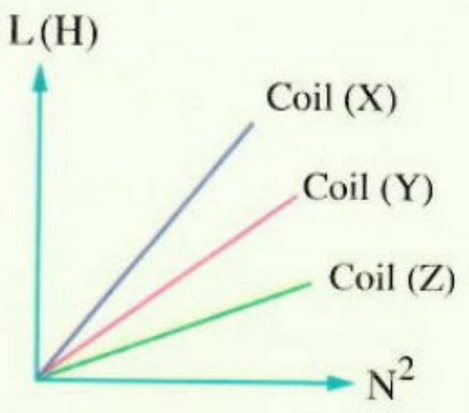



**Graph (1)**




**Graph (2)**

**16** Three solenoids (X), (Y), (Z) having the same cross-section and variable number of turns. The opposite graph represents the relation between the inductance (L) and the square of the number of turns ($N^2$) in each solenoid. Which one from the following is the correct arrangement concerning the lengths ($\ell$) of the solenoids?




(a) $\ell_X > \ell_Y > \ell_Z$ (b) $\ell_Y > \ell_X > \ell_Z$

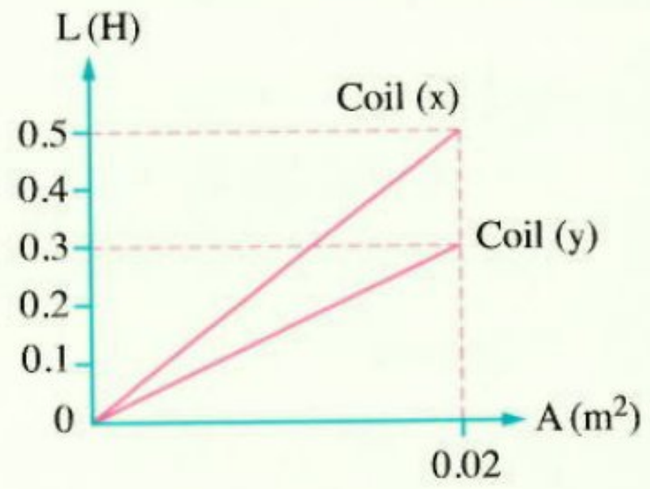
(c) $\ell_Z > \ell_Y > \ell_X$ (d) $\ell_Z > \ell_X > \ell_Y$

*($2^{nd}$ session 21)*



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**17** A group (x) of solenoids, each with a length $\ell$ and a number of turns equal to $N_x$, and another group (y) of solenoids, each with a length of $\frac{\ell}{15}$ and a number of turns equal to $N_y$, and the opposite graph shows the relationship between the self-inductance (L) of each solenoid and the cross-sectional area (A). If all the solenoids have the same magnetic permeability coefficient, then the ratio $\left(\frac{N_y}{N_x}\right)$ equals ..... . *(1st session 24)*




(a) $\frac{3}{5}$ (b) $\frac{2}{5}$ (c) $\frac{1}{5}$ (d) $\frac{4}{5}$

**18** Two solenoids, the first one has a number of turns of (N), a length ($\ell$), and a face area (A), with a self-inductance coefficient of (L), while the second one has a number of turns of (2 N) and the same length ($\ell$). Therefore, the face area of the second solenoid that makes its self-inductance (4 L) is ..... . *(2nd session 24)*

(Where: cores of both solenoids have the same magnetic permeability)

(a) $\frac{1}{4}$ A (b) 2 A (c) $\frac{1}{2}$ A (d) A

**19** A solenoid of insulated wire is 40 cm long and consists of 200 turns, each is of cross-sectional area 4 $cm^2$. The coil is wound around a soft iron cylinder having magnetic permeability of $2 \times 10^{-3}$ T.m/A. As the current in the coil changes at a constant rate, an electromotive force of 2 V is generated by self-induction between its terminals.

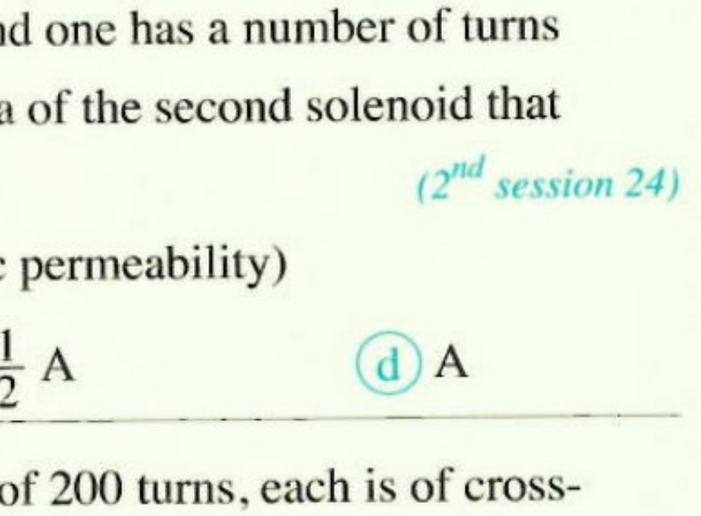
What is the rate of change of the current intensity flowing through the coil? *(1st session 25)*

(a) 4 A/s (b) 500 A/s (c) 25 A/s (d) 0.08 A/s

**20** An inductive coil of length $\ell$ has 800 turns, each of cross-sectional area 20 $cm^2$. The graph represents the relationship between the induced electromotive force (emf) across the coil and the rate of change of current intensity $\left(\frac{\Delta I}{\Delta t}\right)$ passing through it.

What is the coil length ($\ell$)?

(Assuming that: $\mu = 4\pi \times 10^{-7}$ Wb/A.m)




(a) 40 cm (b) 30 cm

(c) 10 cm (d) 20 cm

*(2nd session 25)*




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21 Electric magnet has a coil of resistance 2 Ω and self inductance 2 H is connected to a switch and a battery in a closed electric circuit, on switching the circuit off, the electric current through the circuit vanishes through a time of 0.1 second so that an e.m.f of 150 volts is induced between the terminals of the coil. *(2nd session 23)*

**Calculate:**

**(a)** The intensity of the electric current that passes in the coil before the circuit is opened.

**(b)** The potential difference between the terminal of the coil before the circuit is opened.

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